UNIT I : Introduction to Finite Element Methods

1.1 What is the Finite Element Method?

FEM is a **numerical technique** to find approximate solutions of differential / integral equations that describe physical **field problems** (displacement, temperature, pressure, potential ...) on complicated geometries, loads and boundary conditions where exact (analytical) solutions are not possible.

- The structure (continuum) is divided into many small pieces called **finite elements**, joined at points called **nodes**.
- Over each element the unknown field is approximated by a simple polynomial (interpolation / shape functions).
- Element equations are written, **assembled** into global equations and solved for nodal unknowns.

$$[K]\{Q\} = \{F\}$$

where $[K]$ = global stiffness matrix, $\{Q\}$ = nodal displacement vector, $\{F\}$ = global load vector.

Steps in FEM (general procedure)

1. **Discretisation** : divide the body into elements (type, size, number).
2. Select **displacement (interpolation) function** $u = [N]\{q\}$.
3. Derive **element stiffness matrix** $[k]$ and element load vector $\{f\}$.
4. **Assemble** element equations to get global equations $[K]\{Q\} = \{F\}$.
5. Apply **boundary conditions** (supports) and solve for $\{Q\}$.
6. Post-process : strains $\boldsymbol{\varepsilon} = [B]\{q\}$, stresses $\boldsymbol{\sigma} = [D]\{\boldsymbol{\varepsilon}\}$, reactions.

Applications of FEM

- **Structural** : stress analysis of beams, trusses, frames, plates, shells, machine parts, bridges, aircraft & automobile bodies.
- **Thermal** : conduction, convection in fins, walls, engine blocks, heat exchangers.
- **Fluid flow** : seepage, potential flow, lubrication, aerodynamics.
- **Dynamics** : vibration, modal analysis, crash and impact. **Others** : electromagnetics, biomechanics, acoustics.

Advantages: *Advantages : handles irregular geometry, any loading, mixed materials, any boundary condition, gives stress at every point. Disadvantage : results are approximate; needs a computer and engineering judgement (mesh quality).*

1.2 Stress and Equilibrium

Consider a small element of a body. The internal forces per unit area are ***stresses****. In 3-D there are 6 independent stress components (because of complementary shear,* $\tau_{xy}=\tau_{yx}$ *etc.).*

3-D: 6 components

$\{\sigma\}=[\sigma x\ \sigma y\ \sigma z\ \tau xy\ \tau yz\ \tau zx]^T$

2-D stress element

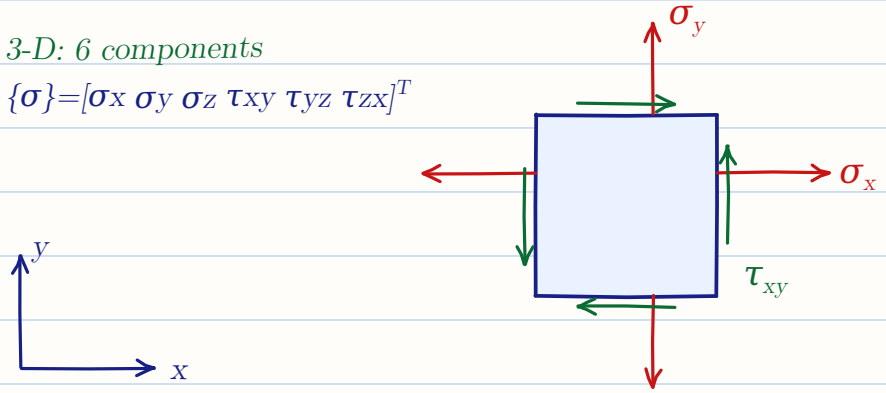


Fig 1.1 Stress components on an element

$$\{\sigma\} = [\sigma_x\ \sigma_y\ \sigma_z\ \tau_{yz}\ \tau_{xz}\ \tau_{xy}]^T$$

Equilibrium equations (3-D)

Take an elemental volume dx dy dz with body force per unit volume X, Y, Z (e.g. weight, centrifugal force). Sum of forces = 0 in each direction :

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + \partial\tau_{xz}/\partial z + X = 0$$

$$\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + \partial\tau_{yz}/\partial z + Y = 0$$

$$\partial\tau_{xz}/\partial x + \partial\tau_{yz}/\partial y + \partial\sigma_z/\partial z + Z = 0$$

For ***2-D*** *problems :* $\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + X = 0$ *and* $\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + Y = 0$.

For ***1-D*** *:* $d\sigma/dx + f = 0$.

Boundary conditions

- ***Essential (geometric / Dirichlet)*** *BC : displacement is specified, e.g.* $u = 0$ *at a fixed support. Must be satisfied by the trial function.*
- ***Natural (force / Neumann)*** *BC : traction / force specified on the boundary, e.g.* $\sigma A = P$ *at a free end. Automatically satisfied in the weak (energy) form.*

Traction on an inclined boundary with direction cosines (n_x, n_y) *:* $T_x = \sigma_x n_x + \tau_{xy} n_y$, $T_y = \tau_{xy} n_x + \sigma_y n_y$.

1.3 Strain - Displacement Relations

Let u, v, w be the displacements in x, y, z. For small deformations (linear theory) :

$$\{\varepsilon\} = \begin{bmatrix} \partial u/\partial x \\ \partial v/\partial y \\ \partial w/\partial z \\ \partial v/\partial z + \partial w/\partial y \\ \partial u/\partial z + \partial w/\partial x \\ \partial u/\partial y + \partial v/\partial x \end{bmatrix}$$

i.e. $\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\varepsilon_z = \partial w/\partial z$ *(normal strains) and* γ_{yz}, γ_{xz}, γ_{xy} *(engineering shear strains).*

2-D *:* $\{\varepsilon\} = [\varepsilon_x \ \varepsilon_y \ \gamma_{xy}]^T = [\partial u/\partial x, \partial v/\partial y, \partial u/\partial y + \partial v/\partial x]^T$. ***1-D*** *:* $\varepsilon = du/dx$.

In FEM : $u = [N]\{q\} \Rightarrow \boldsymbol{\{\varepsilon\} = [B]\{q\}}$, *where [B] = strain-displacement matrix (derivatives of shape functions).*

1.4 Stress - Strain Relations (Hooke's law)

For a linear elastic isotropic material $\{\sigma\} = [D]\{\varepsilon\}$, *E = Young's modulus,* ν *= Poisson's ratio,* $G = E/2(1+\nu)$.

3-D elasticity matrix

$$[D] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-2\nu)/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-2\nu)/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & (1-2\nu)/2 \end{bmatrix}$$

2-D elasticity matrices

$$\text{Plane STRESS } [D] = \frac{E}{(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}$$

$$\text{Plane STRAIN } [D] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & (1-2\nu)/2 \end{bmatrix}$$

- ***Plane stress*** *: thin plate loaded in its plane,* $\sigma_z = \tau_{xz} = \tau_{yz} = 0$ *(but* $\varepsilon_z \neq 0$*). e.g. thin plates, discs.*
- ***Plane strain*** *: long body with uniform section & load,* $\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$ *(but* $\sigma_z \neq 0$*). e.g. dam, tunnel, long cylinder.*

1-D *:* $\sigma = E\,\varepsilon$*. Plane strain is obtained from plane stress by replacing* $E \rightarrow E/(1-\nu^2)$ *and* $\nu \rightarrow \nu/(1-\nu)$*.*

1.5 Potential Energy and Equilibrium

Total potential energy *Π = Strain energy (U) − Work done by external loads (W).*

$$\Pi = U - W = \tfrac{1}{2} \int \{\varepsilon\}^T\{\sigma\}\, dV - \int \{u\}^T\{X\}\, dV - \int \{u\}^T\{T\}\, dS - \Sigma\, \{u_i\}^T\{P_i\}$$

Principle: *Principle of Minimum Potential Energy : Among all admissible displacement fields (satisfying the essential BC) the one that satisfies equilibrium makes* Π *a minimum* $\Rightarrow \delta\Pi = 0$*, i.e.* $\partial\Pi/\partial q_i = 0$*.*

For a discretised system $\Pi = \tfrac{1}{2}\{Q\}^T[K]\{Q\} - \{Q\}^T\{F\}$*. Setting* $\partial\Pi/\partial\{Q\} = 0$ *gives* $\boldsymbol{[K]\{Q\} = \{F\}}$*, the equilibrium equation of FEM.*

Rayleigh - Ritz method

Assume the displacement as a finite series with unknown constants, $u = a_1\varphi_1(x) + a_2\varphi_2(x) + \ldots + a_n\varphi_n(x)$, each φ_i satisfying the essential BC. Substitute in Π and apply $\partial\Pi/\partial a_i = 0$ to get n algebraic equations.

Example : Bar of length L, constant AE, fixed at $x = 0$, axial load P at $x = L$. Take $u = a\,x$:

$$\Pi = \tfrac{1}{2}\int_0^L AE\,(du/dx)^2\,dx - P\,u(L) = \tfrac{1}{2}\,AE\,a^2\,L - P\,a\,L$$

$$\partial\Pi/\partial a = AE\,a\,L - P\,L = 0 \Rightarrow a = P/AE,\ u = P\,x/AE \text{ (exact)}$$

- Limitation : trial function must be written for the **whole** domain, difficult for complex shapes. ***FEM = Rayleigh-Ritz applied piece-wise*** on every element with local shape functions.
- Galerkin method (weighted residual) : weight functions = shape functions; gives the same equations for structural problems.

1.6 Formulation of Finite Element Equations

Within an element $\{u\} = [N]\{q\}$, $\{\varepsilon\} = [B]\{q\}$, $\{\sigma\} = [D][B]\{q\}$. Substituting in Π (element level) :

$$\Pi_e = \tfrac{1}{2}\{q\}^T\left[\int [B]^T[D][B]\,dV\right]\{q\} - \{q\}^T\left(\int [N]^T\{X\}\,dV + \int [N]^T\{T\}\,dS + \{P\}\right)$$

$$[k] = \int_V [B]^T\,[D]\,[B]\,dV \quad \{f\} = \int_V [N]^T\{X\}dV + \int_S [N]^T\{T\}dS + \{P\}$$

Minimising Π_e gives the element equation $\mathbf{[k]\{q\} = \{f\}}$. Properties of $[k]$: (i) symmetric, (ii) singular (rigid-body motion possible before supports are applied), (iii) diagonal terms positive, (iv) banded.

1.7 One-Dimensional Problems : the Bar Element

A 1-D bar carries only axial load. Each node has **one** degree of freedom (axial displacement). A 2-noded linear element has 2 dof : $\{q\} = [q_1\ q_2]^T$.

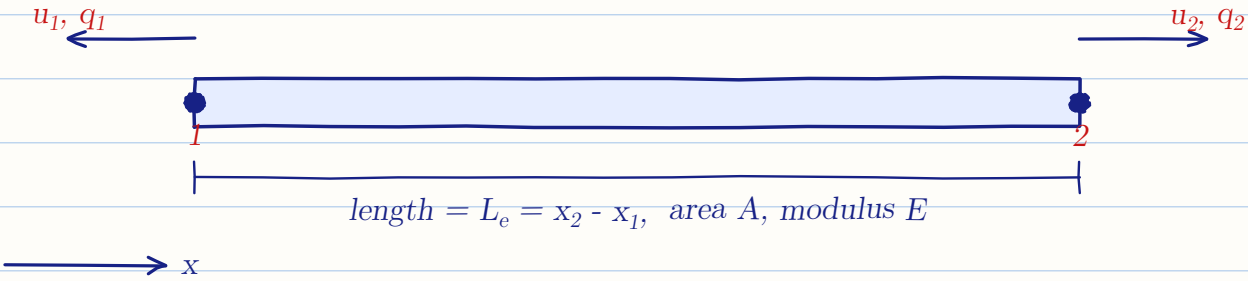


Fig 1.2 Two-noded 1-D bar element

Coordinates and shape functions

Use the ***natural (intrinsic) coordinate*** ξ *that goes from* -1 *at node 1 to* $+1$ *at node 2 :*

$$\xi = 2(x - x_1)/(x_2 - x_1) - 1,\ x = \tfrac{1}{2}(1-\xi)x_1 + \tfrac{1}{2}(1+\xi)x_2$$

$$N_1 = (1 - \xi)/2\ N_2 = (1 + \xi)/2\ u = N_1 q_1 + N_2 q_2$$

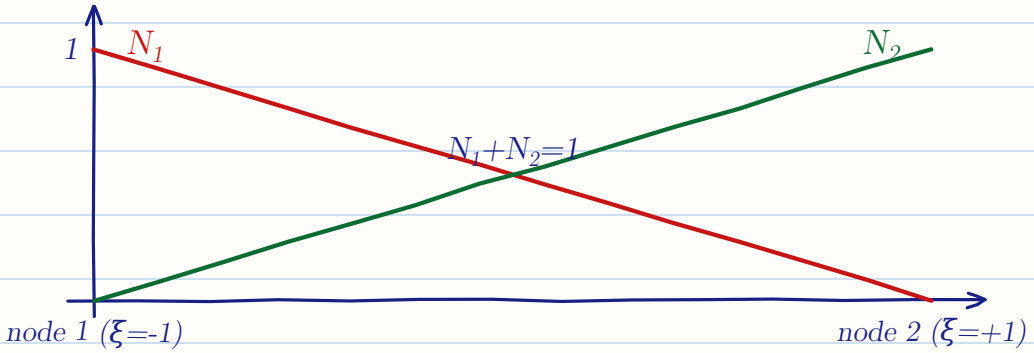


Fig 1.3 Linear shape functions

Properties : $N_i = 1$ *at node* i *and* 0 *at the other node;* $\Sigma N_i = 1$ *at every point (rigid-body motion is represented);* $dx = (L_e/2)\, d\xi$ *(Jacobian* $J = L_e/2$*).*

Strain-displacement matrix and stiffness matrix

$$\varepsilon = du/dx = (du/d\xi)(d\xi/dx) = (2/L_e)\,(du/d\xi) \Rightarrow [B] = (1/L_e)\,[-1\ \ 1]$$

$$[k] = \int [B]^T E\,[B]\,A\,dx = A_e E_e\, L_e/L_e^{\,2}\,[\ \ldots\ ]$$

$$[k] = A_e E_e / L_e \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Meaning: *Physical meaning : the bar behaves as a spring of stiffness* $k = AE/L$*. Column 1 = forces needed for unit displacement of node 1 with node 2 fixed.*

Load vectors

$$\text{Body force } f \text{ per unit volume}: \quad \{f_b\} = A_e L_e f/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Traction } T \text{ per unit area (on surface)}: \quad \{f_T\} = T\, L_e/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Thermal load (rise } \Delta T): \quad \{f_{th}\} = E_e A_e \alpha\, \Delta T \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

1.8 Requirements for Convergence; Interpolation Functions

For the FE solution to converge to the exact solution as the mesh is refined, the displacement (interpolation) function must satisfy :

1. ***Completeness*** : must include ***rigid-body modes*** and ***constant-strain states*** (so that when element size $\rightarrow 0$ the strain becomes uniform).
2. ***Compatibility (conformity)*** : displacement must be continuous across element boundaries (no gaps or overlaps). Elements satisfying it are called conforming.
3. ***Geometric isotropy (invariance)*** : the polynomial should not prefer a direction; the result must not depend on orientation of the axes (use all terms of Pascal triangle symmetrically).
4. Number of terms in polynomial = number of element dof.

Interpolation function (polynomial) is preferred because it is easy to differentiate and integrate and the order can be increased to improve accuracy. A 1-D polynomial : $u = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \ldots$

Pascal's triangle

Used to pick polynomial terms for 2-D elements so that the polynomial stays complete and symmetric in x and y :

constant			1			CST uses 1, x, y
linear			x		y	
quadratic		x^2		xy		y^2 Q4 adds xy
cubic		x^3		x^2y	xy^2	y^3
quartic	x^4		x^3y		x^2y^2	xy^3 y^4

Fig 1.4 Pascal's triangle

- *3 nodes (CST) : $u = \alpha_1 + \alpha_2 x + \alpha_3 y$ (complete linear, 3 terms). 4 nodes (Q4) : add the xy term (bilinear). 6 nodes (LST) : complete quadratic (6 terms).*

1.9 Assembly of Global Stiffness Matrix and Load Vector

*Consider a bar with n elements and n+1 nodes, global dof Q_1 ... Q_{n+1}. Element 1 connects nodes (1,2), element 2 connects (2,3), and so on. The element matrix is added to the global matrix at the rows & columns of its connected nodes (**direct stiffness method**). For 3 elements (4 nodes) :*

$$[K]\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2+k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix}\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{Bmatrix}\begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

where $k_i = A_i E_i / L_i$. The global load vector is assembled in the same way : $F_2 = f_2^{(1)} + f_1^{(2)} +$ (point load at node 2), etc.

Treatment of boundary conditions

Before solving, [K] is singular. Supports must be applied. Two methods :

***(a) Elimination method** : for the dof $Q_r = 0$, strike out row r and column r (reduced system). Solve for the remaining Q. Then support reaction $R_r = [K]_{r,*}\{Q\} - F_r$.*

For a non-zero support displacement $Q_r = a$, move the column terms to the right side : $F_i \leftarrow F_i - K_{ir}\, a$ for all remaining rows.

***(b) Penalty approach** : add a very large number C ($\approx 10^4 \times \max K_{ii}$) to the diagonal K_{rr} and add $C \cdot a$ to F_r. Then solve the full system. Reaction $R_r = -C$*

$(Q_r - a)$.

Solved Example 1 : stepped bar

Bar fixed at node 1, $P = 50$ kN at node 3. $E = 200$ GPa $= 2\times10^5$ N/mm^2. Element 1: A_1=400 mm^2, L_1=300 mm; element 2: A_2=250 mm^2, L_2=300 mm. Find displacements, stresses, reaction.

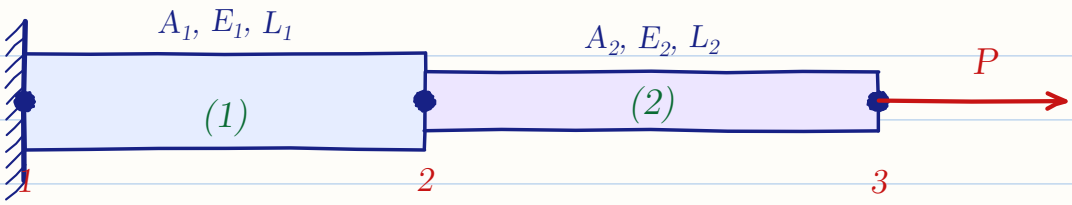


Fig 1.5

$k_1 = A_1E/L_1 = 400\times2\times10^5/300 = 2.667\times10^5$ N/mm ; $k_2 = 250\times2\times10^5/300 = 1.667\times10^5$ N/mm

$$[K]\begin{bmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1+k_2 & -k_2 \\ 0 & -k_2 & k_2 \end{bmatrix}\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{Bmatrix} = [R_1,\ 0,\ 50000]^T$$

Apply $Q_1 = 0$ (strike row 1 & column 1). Reduced equations :

$$\begin{bmatrix} 4.333\times10^5 & -1.667\times10^5 \\ -1.667\times10^5 & 1.667\times10^5 \end{bmatrix}\begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [0,\ 50000]^T$$

Solving : $Q_2 = 0.1875$ mm, $Q_3 = 0.4875$ mm. Stresses : $\sigma_1 = E(Q_2-Q_1)/L_1 = 2\times10^5\times0.1875/300 = 125$ N/mm^2 ; $\sigma_2 = E(Q_3-Q_2)/L_2 = 2\times10^5\times0.3/300 = 200$ N/mm^2.

Reaction $R_1 = -k_1Q_2 = -2.667\times10^5\times0.1875 = -50\,000$ N (balances P). Check : $\sigma = P/A$ gives 125 and 200 OK.

1.10 Quadratic Shape Functions (3-noded bar)

A 3-noded element (nodes 1, 2 at the ends, node 3 at mid-length) gives a **parabolic** displacement and linearly varying strain : $u = \alpha_1 + \alpha_2\xi + \alpha_3\xi^2$. Using the condition $N_i(\text{node } j) = \delta_{ij}$:

$$N_1 = \xi(\xi - 1)/2 \quad N_2 = \xi(\xi + 1)/2 \quad N_3 = 1 - \xi^2$$

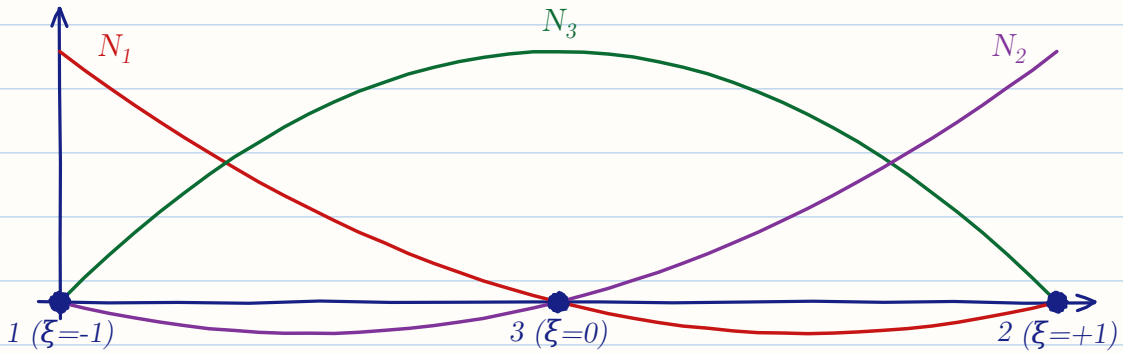


Fig 1.6 Quadratic shape functions (ξ = -1, +1, 0)

$[B] = (2/L_e)\ [\ dN_1/d\xi\ dN_2/d\xi\ dN_3/d\xi\ ] = (2/L_e)\ [\ (2\xi-1)/2\ ,\ (2\xi+1)/2\ ,\ -2\xi\ ]$.

Using $[k] = \int_{-1}^{1} [B]^T E[B]\ A\ (L_e/2)\ d\xi$:

$$[k] = A_e E_e/(3L_e) \begin{bmatrix} 7 & 1 & -8 \\ 1 & 7 & -8 \\ -8 & -8 & 16 \end{bmatrix}$$

$$\text{Body-force vector } \{f\} = f\ A_e\ L_e/6 \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$$

Check : the sum of each row of $[k]$ is zero (rigid-body motion), and body-force weights 1/6, 1/6, 4/6 add to 1 (Simpson's rule).

Compare: Linear element : constant strain, needs many elements. Quadratic element : linear strain, better accuracy with fewer elements (same dof count gives higher accuracy).

1.11 More Solved Problems

Example 2 : Penalty approach (spring-like support)

Same stepped bar as Example 1. Use penalty method for $Q_1 = 0$. Choose $C = 10^4 \times \max K_{ii} \approx 4.3\times10^9$ N/mm.

$$\begin{bmatrix} 4.333\times10^5+C & -2.667\times10^5 & 0 \\ -2.667\times10^5 & 4.333\times10^5 & -1.667\times10^5 \\ 0 & -1.667\times10^5 & 1.667\times10^5 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{Bmatrix} = [0,\ 0,\ 50000]^T$$

(Here $K_{11} = k_1 + C$.) Because C is huge, $Q_1 \approx 0$ (about 10^{-5} mm) and the remaining equations reduce to those of the elimination method, giving $Q_2 = 0.1875$ mm, $Q_3 = 0.4875$ mm. Reaction $R_1 = -C\,Q_1 \approx -50\,000$ N.

Remark: Penalty method keeps the matrix size unchanged and is easy to program; but too large C causes round-off errors, too small C gives an inexact constraint.

Example 3 : Bar under self weight (body force)

A uniform bar of length $L = 2$ m, $A = 100$ mm^2, $E = 2\times10^5$ N/mm^2, weight density $\rho g = 7.85\times10^{-5}$ N/mm^3, hangs from a fixed support at the top. Model with 2 elements of $L_e = 1000$ mm.

Body-force vector of each element : $f\,A\,L_e/2 = 7.85\times10^{-5}\times100\times1000/2 = 3.925$ N at each end node. Global load : $F_1 = R_1 + 3.925$, $F_2 = 7.85$, $F_3 = 3.925$.

$k = AE/L_e = 100\times2\times10^5/1000 = 2\times10^4$ N/mm. Apply $Q_1 = 0$:

$$\begin{bmatrix} 2k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [7.85,\ 3.925]^T$$

Adding the equations: $k\,Q_2 = 11.775 \Rightarrow Q_2 = 5.89\times10^{-4}$ mm; then $k(Q_3-Q_2) = 3.925 \Rightarrow Q_3 = 7.85\times10^{-4}$ mm. Exact value at the tip $\rho g L^2/2E = 7.85\times10^{-4}$ mm, hence **exact at the nodes** (for linear elements and uniform loading).

Example 4 : Thermal stress in a bar

Steel bar ($E = 2\times10^5$ N/mm^2, $\alpha = 12\times10^{-6}$/°C, $A = 500$ mm^2, $L = 400$ mm) is rigidly fixed at both ends and heated by $\Delta T = 50$ °C. Find stress.

One element. Thermal load $\{f_{th}\} = EA\alpha\Delta T\,[-1,\ 1]^T = 2\times10^5\times500\times12\times10^{-6}\times50 = 60\,000$ N. Both ends fixed ($Q_1 = Q_2 = 0$), so all displacements are zero and the strain due to loading is zero. Mechanical strain $= -\alpha\Delta T$:

$\sigma = E(\varepsilon - \alpha\Delta T) = 2\times10^5(0 - 12\times10^{-6}\times50) = -120\ N/mm^2$ (compressive)

Example 5 : Quadratic element

A 3-noded bar ($L_e = 100$ mm, $A = 200\ mm^2$, $E = 2\times10^5\ N/mm^2$) is fixed at node 1; a load of 10 kN acts at node 2 (end). Find the displacement at the mid-node. $AE/(3L_e) = 200\times2\times10^5/300 = 1.333\times10^5$. With $Q_1 = 0$ the remaining equations (nodes 2, 3) are :

$$1.333\times10^5 \begin{bmatrix} 7 & -8 \\ -8 & 16 \end{bmatrix} \begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [10000,\ 0]^T$$

From row 2 : $Q_3 = Q_2/2$; row 1 : $7Q_2 - 4Q_2 = 3Q_2 = 10000/1.333\times10^5 = 0.075$ $\Rightarrow Q_2 = 0.025$ mm, $Q_3 = 0.0125$ mm. This equals the exact $PL/AE = 10^4\times100/(200\times2\times10^5) = 0.025$ mm.

Galerkin (weighted residual) form for the 1-D bar

Governing equation : $d/dx(AE\ du/dx) + f = 0$. Multiply by a weight function w ($= \delta u$, same as shape function) and integrate over the length; integrate by parts :

$$\int (dw/dx)\ AE\ (du/dx)\ dx = \int w\ f\ dx + [w\ AE\ du/dx]_{boundary}$$

Substituting $u = [N]\{q\}$, $w = [N]\{\delta q\}$ gives $\int [B]^T AE[B]\ dx\ \{q\} = \int [N]^T f\ dx +$ boundary forces, i.e. exactly $[k]\{q\} = \{f\}$. The natural BC appears automatically in the boundary term, hence it is called the **weak form**.

Shape function derivation (step by step, 2-noded)

1. Assume $u = \alpha_1 + \alpha_2 x$. 2. Apply $u(x_1) = q_1$, $u(x_2) = q_2$ and solve for α_1, α_2.
3. $\alpha_2 = (q_2 - q_1)/(x_2 - x_1)$; $\alpha_1 = (q_1x_2 - q_2x_1)/(x_2 - x_1)$.
4. Substitute back : $u = [(x_2-x)/L_e]\ q_1 + [(x-x_1)/L_e]\ q_2 \Rightarrow N_1 = (x_2-x)/L_e$, $N_2 = (x-x_1)/L_e$ (same as $(1\mp\xi)/2$).

Quick revision / expected questions

- Define FEM; list steps and applications. Write equilibrium equations in 2-D / 3-D.
- Derive strain-displacement and [D] for plane stress and plane strain. Differences?
- State principle of minimum potential energy; explain Rayleigh-Ritz with example.
- Derive shape functions, [B] and [k] of 1-D bar; assemble for a stepped bar;

elimination vs penalty.

- Convergence criteria; Pascal's triangle; derive quadratic shape functions.

UNIT II : 1-D Analysis of Trusses and Beams

2.1 Trusses : introduction

A **truss** is a structure of straight slender members joined at their ends by frictionless pins. Loads act only at the joints, so every member carries **axial force only** (tension or compression) - no bending.

- Planar truss : every joint has **2 dof** (u, v) in global X-Y. A bar element has 2 nodes ⇒ **4 dof**.
- Difference from 1-D bar : members are **inclined** at different angles, so the element matrix must be **transformed** from local axis (along the member) to global axes.
- Assumptions : pin joints, loads at joints only, linear elastic material, small displacement, weight of members neglected.

2.2 Stiffness matrix of a 1-D truss element

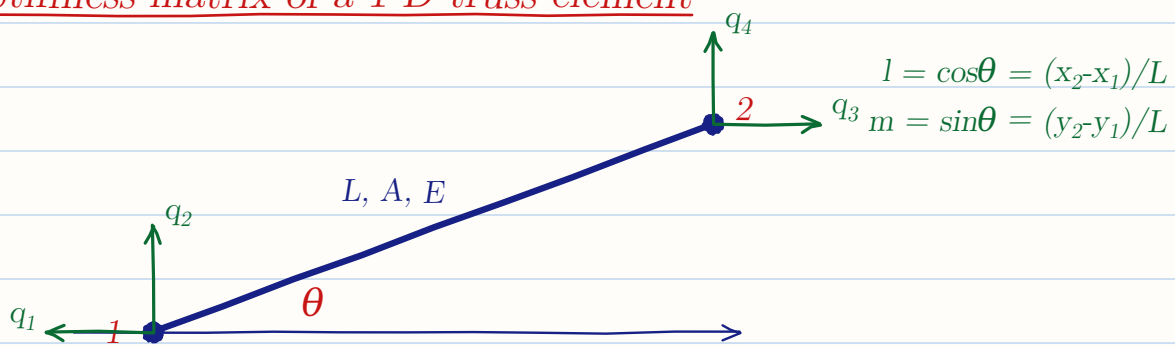


Fig 2.1 Truss element in global co-ordinates

In **local** co-ordinates (x' along the member) the element has 2 dof q'_1, q'_2 and the bar stiffness :

$$[k'] = AE/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Relation between local and global displacements : $u'_1 = q_1 \cos\theta + q_2 \sin\theta = l\, q_1 + m\, q_2$, and $u'_2 = l\, q_3 + m\, q_4$. In matrix form :

$$\{q'\} = [L]\{q\} = \begin{bmatrix} l & m & 0 & 0 \\ 0 & 0 & l & m \end{bmatrix}$$

Strain energy is invariant : $\frac{1}{2}\{q'\}^T[k']\{q'\} = \frac{1}{2}\{q\}^T[k]\{q\} \Rightarrow \mathbf{[k] = [L]^T[k'][L]}$.

Carrying out the multiplication :

$$[k] = AE/L \begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}$$

$l = \cos\theta = (x_2 - x_1)/L \; m = \sin\theta = (y_2 - y_1)/L \; L = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Remarks: *[k] is symmetric and singular; each row sums to zero; for a horizontal member ($\theta = 0$) it reduces to the 1-D bar matrix. Dof numbering for a node i : (2i−1, 2i). Angle θ is measured counter-clockwise from +X to the direction node 1 → node 2.*

Stress calculation

Stress in the member $\sigma = E\,\varepsilon = E\,(u'_2 - u'_1)/L$:

$$\sigma = (E/L)\ [\ -l\ -m\ l\ m\ ]\ \{q_1\ q_2\ q_3\ q_4\}^T$$

Positive σ = tension, negative = compression. Member axial force $F = \sigma\ A$. Support reactions are obtained from $\{R\} = [K]\{Q\} - \{F\}$, using the rows of the restrained dof.

Procedure for truss analysis

1. Number nodes & elements; write coordinates, A, E of each member.
2. Compute L, l, m for every element; form element [k] (4×4).
3. Assemble global [K] (2 × number of nodes) using dof (2i−1, 2i) of each node.
4. Apply BC (elimination / penalty), solve $[K]\{Q\} = \{F\}$.
5. Find stresses $\sigma = (E/L)[-l\ -m\ l\ m]\{q\}$; reactions from the restrained rows.

2.3 Solved Problem 1 : three-bar truss

Triangular truss : nodes 1 (0,0), 2 (800,0), 3 (400,300) mm. Members (1): 1-2, (2): 2-3, (3): 1-3, all $A = 500\ mm^2$, $E = 2\times10^5\ N/mm^2$. Node 1 pinned, node 2 on a roller ($v_2 = 0$). Vertical load $P = 10\ kN$ (downward) at node 3. Find displacements, stresses, reactions.

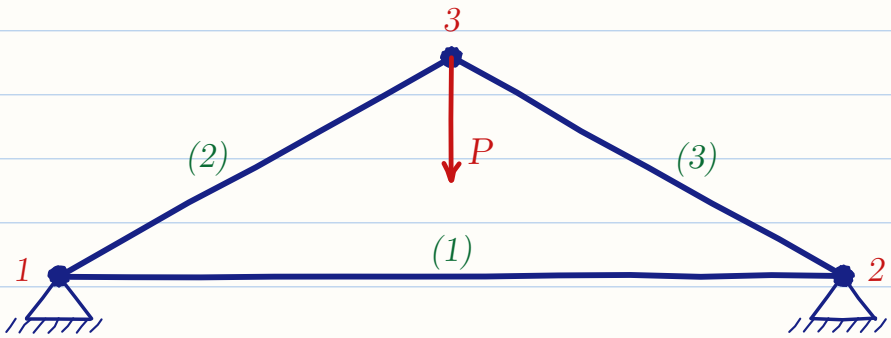


Fig 2.2 (members 1-2 bottom, 2-3 right, 1-3 left)

Step 1 : element data *($AE = 10^8$ N)*

Elem	Nodes	L (mm)	l	m	AE/L (N/mm)
1	1-2	800	1.00	0.00	125000
2	2-3	500	-0.80	0.60	200000
3	1-3	500	0.80	0.60	200000

Step 2 : assemble. *Global dof order : Q_1=u_1, Q_2=v_1, Q_3=u_2, Q_4=v_2, Q_5=u_3, Q_6=v_3. Element dof : (1) = (1,2,3,4), (2) = (3,4,5,6), (3) = (1,2,5,6). Boundary conditions : $Q_1 = Q_2 = Q_4 = 0$, so keep rows/columns 3, 5, 6. Loads : $F_3 = 0$, $F_5 = 0$, $F_6 = -10\,000$ N.*

$$\begin{bmatrix} 253000 & -128000 & 96000 \\ -128000 & 256000 & 0 \\ 96000 & 0 & 144000 \end{bmatrix} \begin{Bmatrix} Q_3 \\ Q_5 \\ Q_6 \end{Bmatrix} = [0,\ 0,\ -10000]^T$$

Step 3 : solve *: $Q_3 = u_2 = 0.0533$ mm, $Q_5 = u_3 = 0.0267$ mm, $Q_6 = v_3 = -0.1050$ mm.*

Step 4 : stresses σ = (E/L)[−l −m l m]{q} :

- *Element 1 (1-2) : σ = 13.33 N/mm^2 (tension), force = 6667 N*
- *Element 2 (2-3) : σ = -16.67 N/mm^2 (compression), force = -8333 N*
- *Element 3 (1-3) : σ = -16.67 N/mm^2 (compression), force = -8333 N*

Step 5 : reactions *: $R_{1x} = 0$ N, $R_{1y} = 5000$ N, $R_{2y} = 5000$ N. Check $\Sigma F_y = 5000 + 5000 = 10\,000$ N, $\Sigma F_x = 0 \approx 0$ - equilibrium satisfied.*

2.4 Solved Problem 2 : two-bar bracket

Two bars hold a load $P = 10$ kN at joint 3. Nodes 1 (0,300), 2 (0,−300) (walls, fixed), 3 (400,0). $A = 500$ mm^2, $E = 2\times10^5$ N/mm^2. Only joint 3 can move : dof Q_5 (u_3), Q_6 (v_3).

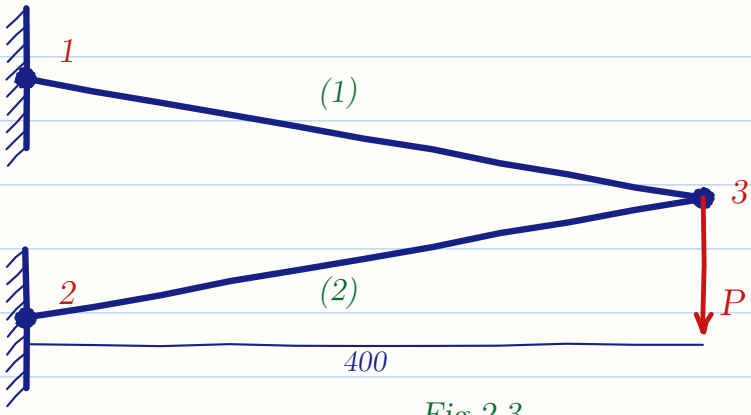


Fig 2.3

Elem	Nodes	L (mm)	l	m	AE/L (N/mm)
1	1-3	500	0.80	-0.60	200000
2	2-3	500	0.80	0.60	200000

For each element only the lower-right 2×2 block (node 3) is needed : $k_{ij} = (AE/L) \times [l^2\ lm\ ;\ lm\ m^2]$.

$$\begin{bmatrix} 256000 & 0 \\ 0 & 144000 \end{bmatrix} \begin{Bmatrix} u_3 \\ v_3 \end{Bmatrix} = [0,\ -10000]^T$$

Result : $u_3 = 0.0000$ mm, $v_3 = -0.0694$ mm. Stresses : $\sigma_1 = 16.67$ N/mm^2, $\sigma_2 = -16.67$ N/mm^2. (Member 1, the upper bar, is in tension; member 2, the lower bar, in compression - matches statics : $F = P/(2\sin\alpha) = 8333$ N.)

2.5 Analysis of Beams

A **beam** carries transverse loads and resists them by bending. Each node has **2 dof** : transverse deflection v and slope (rotation) $\theta = dv/dx$. A 2-noded beam element therefore has **4 dof**.

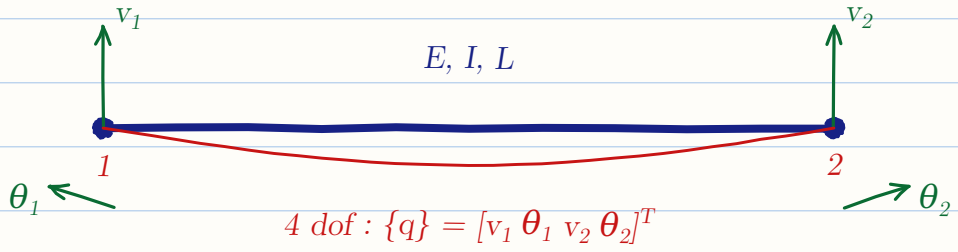


Fig 2.4 Beam element

Beam theory (Euler-Bernoulli) : $M = EI\, d^2v/dx^2$, $V = dM/dx$, $EI\, d^4v/dx^4 = q(x)$. Strain energy $U = \frac{1}{2} \int EI\, (d^2v/dx^2)^2\, dx$. Since second derivatives appear, the shape functions must be **C^1 continuous** (both v and dv/dx continuous between elements) ⇒ **Hermite cubic** polynomials.

Hermite shape functions

Four dof ⇒ cubic : $v = a_0 + a_1\xi + a_2\xi^2 + a_3\xi^3$ with $-1 \le \xi \le 1$, $x = (x_1+x_2)/2 + (L/2)\xi$ and $dv/dx = (2/L)\, dv/d\xi$. Imposing $v(-1) = v_1$, $v'(-1) = \theta_1$, $v(1) = v_2$, $v'(1) = \theta_2$ gives :

$$H_1 = (1/4)(1-\xi)^2(2+\xi) \quad H_2 = (L/8)(1-\xi)^2(1+\xi)$$

$$H_3 = (1/4)(1+\xi)^2(2-\xi) \quad H_4 = (L/8)(1+\xi)^2(\xi-1)$$

$$v(\xi) = H_1v_1 + H_2\theta_1 + H_3v_2 + H_4\theta_2 = [H]\{q\}$$

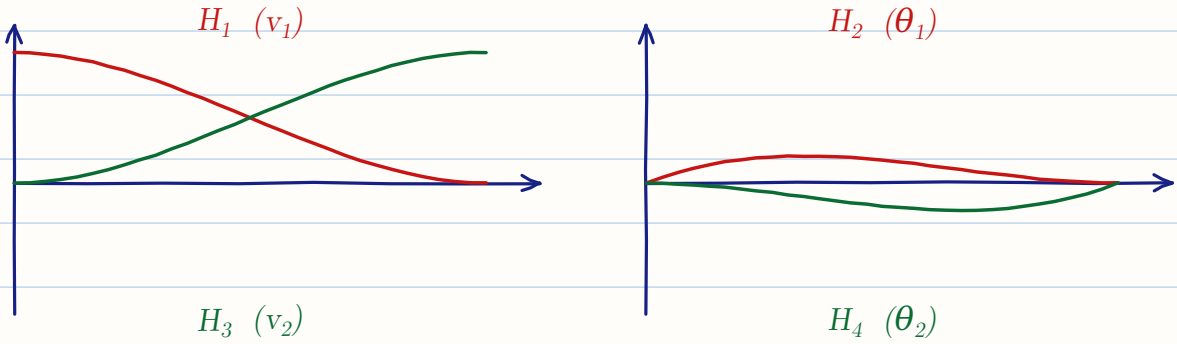


Fig 2.5 Hermite shape functions

Properties : H_1=1 at node 1 and H_1=H_1'=0 at node 2 (similarly H_3); H_2'=1 at node 1 with zero value and slope elsewhere (similarly H_4); $H_1+H_3 = 1$ (rigid-body translation).

Element stiffness matrix

Curvature $d^2v/dx^2 = (4/L^2)\, d^2v/d\xi^2 = [B]\{q\}$, $[B] = (1/L^2)\,[\,6\xi\,,\, L(3\xi-1)\,,\, -6\xi\,,\, L(3\xi+1)\,]$.

$$[k] = \int_0^L [B]^T\, EI\, [B]\, dx = (EI\, L/2) \int_{-1}^{1} [B]^T [B]\, d\xi$$

$$[k] = EI/L^3 \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

DOF order $\{v_1, \theta_1, v_2, \theta_2\}$. $[k]$ is symmetric, singular (2 rigid-body modes : translation and rotation).

Load vector (consistent nodal loads)

$\{f\} = \int [H]^T\, q\, dx$. For a **uniformly distributed load** q (positive in the $+v$ direction) :

$$\{f\} = qL/2 \begin{bmatrix} 1 \\ L/6 \\ 1 \\ -L/6 \end{bmatrix}$$

i.e. end forces $qL/2$ and fixed-end moments $\pm\, qL^2/12$ (same as fixed-end reactions). For a **point load** P at a node, it is added directly to the v dof; an applied couple M_0 to the θ dof. For a point load inside an element, use fixed-end values $P/2$, $PL/8$, $P/2$, $-PL/8$ (load at the middle).

After solving : bending moment at any section $M = EI\, [B]\{q\}$; at node 1 ($\xi=-1$) and node 2 ($\xi=+1$).

2.6 Solved Problems on Beams

Problem 3 : cantilever with end load

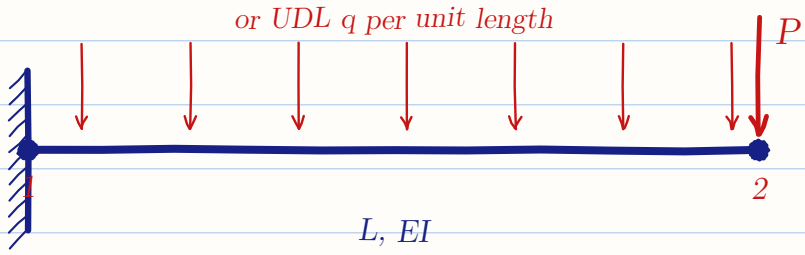


Fig 2.6

One element, $L = 1000$ mm, $E = 2\times10^5$ N/mm^2, $I = 10^7$ mm^4 ($EI = 2\times10^{12}$ N · mm^2), $P = 1000$ N downward at the free end. BC : $v_1 = \theta_1 = 0$. Retain rows 3, 4 :

$$EI/L^3 \begin{bmatrix} 12 & -6L \\ -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} v_2 \\ \theta_2 \end{Bmatrix} = [-P\ ,\ 0]^T$$

Solving : $v_2 = -PL^3/(3EI) = -1000\times10^9/(3\times2\times10^{12}) = -0.1667$ mm (downward), $\theta_2 = -PL^2/(2EI) = -0.00025$ rad. These agree with the **exact** beam-theory results. Reactions from rows 1-2 : $R_1 = +P = 1000$ N upward and $M_1 = PL = 10^6$ N · mm (anticlockwise).

Problem 4 : cantilever with UDL (one element)

Load vector at node 2 : $F = [-qL/2,\ +qL^2/12]^T$ (downward q). Solve EI/L^3 [12 $-6L$; $-6L$ $4L^2$]$\{v_2, \theta_2\} = F$:

$$v_2 = -qL^4/(8EI)\ \theta_2 = -qL^3/(6EI) \text{ (both exact)}$$

Problem 5 : simply supported beam with central load (2 elements)

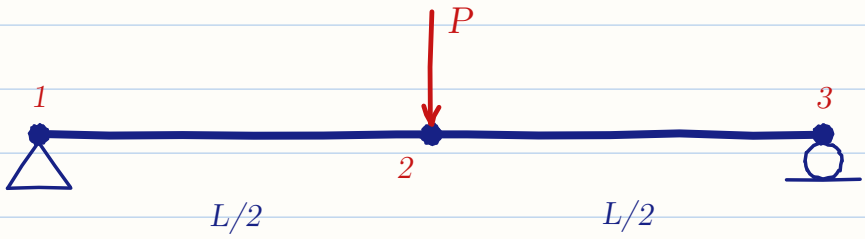


Fig 2.7

Two elements of length $a = L/2$. Nodes 1, 2, 3. BC : $v_1 = v_3 = 0$. Free dof : θ_1, v_2, θ_2, θ_3. Load P downward at node 2 : $F_{v2} = -P$. By symmetry $\theta_2 = 0$ and $\theta_3 = -\theta_1$. The reduced equations give :

$$v_2 = -P(2a)^3/(48EI) = -PL^3/(48EI)\ \theta_1 = -PL^2/(16EI) \text{ (exact)}$$

Moment at the centre $M = EI\,v'' = PL/4$ (obtained from [B] at $\xi = -1$ of element 2). Because the nodal values are exact for beams loaded at nodes, only the interior curve (cubic) is approximate for distributed loads.

Summary: Beam & truss formulas to remember : truss $[k] = AE/L\ [l^2\ lm\ ...]$ $\sigma = (E/L)[-l\ -m\ l\ m]\{q\}$. beam $[k] = EI/L^3\ [12\ 6L\ ...]$, UDL load vector $qL/2\ [1,\ L/6,\ 1,\ -L/6]$, Hermite cubics H_1 to H_4 (C^1 continuity).

2.7 Additional Worked Problems

Direction cosines : quick table

θ (deg)	0	30	45	60	90	120
$l = \cos\theta$	1	0.866	0.707	0.5	0	−0.5
$m = \sin\theta$	0	0.5	0.707	0.866	1	0.866

Always find θ from the coordinates (node 1 → node 2), not from the picture. Swapping node numbers changes the signs of l and m but gives the same $[k]$ and the same stress.

Problem 5 (numerical) : simply supported beam, P = 10 kN at centre

$L = 2000$ mm (two elements of $a = 1000$ mm), $EI = 2\times10^{12}$ N · mm^2. Free dof : $\theta_1,\ v_2,\ \theta_2,\ \theta_3$. Solution of the reduced 4×4 system :

θ_1 (rad)	v_2 (mm)	θ_2 (rad)	θ_3 (rad)
-0.001250	-0.8333	0.000000	0.001250

Exact : $v_2 = -PL^3/48EI = -10^4\times8\times10^9/(48\times2\times10^{12}) = -0.8333$ mm ; $\theta_1 = -PL^2/16EI = -0.00125$ rad. Reactions : $R_1 = R_3 = P/2 = 5000$ N. Maximum bending moment $PL/4 = 5\times10^6$ N · mm.

Problem 6 : three-bar truss meeting at one joint

Three bars ($A = 500$ mm^2, $E = 2\times10^5$ N/mm^2) join at node 4 (400,0). Their other ends are fixed at 1 (0,300), 2 (0,−300), 3 (0,0). Joint 4 carries 5 kN horizontally (+X) and 10 kN downward. Only dof $Q_7 = u_4$, $Q_8 = v_4$ are free.

Elem	Nodes	L	l	m	AE/L
1	1-4	500	0.80	-0.60	200000
2	2-4	500	0.80	0.60	200000
3	3-4	400	1.00	0.00	250000

$$\begin{bmatrix} 506000 & 0 \\ 0 & 144000 \end{bmatrix} \begin{Bmatrix} u_4 \\ v_4 \end{Bmatrix} = [5000,\ -10000]^T$$

Solution : u_4 = 0.0099 mm, v_4 = -0.0694 mm. Stresses : σ_1 = 19.83, σ_2 = -13.50, σ_3 = 4.94 N/mm^2. The horizontal bar 3 takes the horizontal load; bars 1 and 2 take the vertical load (statically indeterminate structure - FEM handles it without extra work).

Beam [B] matrix and a typical stiffness term

Differentiate the Hermite functions twice with respect to ξ and multiply by $(4/L^2)$:

	H_1	H_2	H_3	H_4
d^2H/dx^2	$6\xi/L^2$	$(3\xi-1)/L$	$-6\xi/L^2$	$(3\xi+1)/L$

Example : $k_{11} = EI\ (L/2) \int_{-1}^{1} (6\xi/L^2)^2\ d\xi = EI\ (L/2)(36/L^4)(2/3) = 12EI/L^3$; $k_{22} = EI\ (L/2)(1/L^2) \int (3\xi-1)^2\ d\xi = EI/(2L) \times 8 = 4EI/L$.

Problem 7 : propped cantilever with UDL (1 element)

Fixed at node 1 ($v_1 = \theta_1 = 0$), roller at node 2 ($v_2 = 0$). Only θ_2 is free : $(4EI/L)\ \theta_2 = F_{\theta 2} = +qL^2/12 \Rightarrow \theta_2 = qL^3/(48EI)$. Reaction at the roller $R_2 = (EI/L^3)(-6L\ \theta_2) + qL/2 = 3qL/8$ (exact; the fixed end then carries $5qL/8$).

Problem 8 : simply supported beam with UDL (1 element)

$$EI/L^3 \begin{bmatrix} 4L^2 & 2L^2 \\ 2L^2 & 4L^2 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = [-qL^2/12\ ,\ +qL^2/12]^T$$

$\theta_1 = -qL^3/(24EI)$, $\theta_2 = +qL^3/(24EI)$ (exact end slopes). Mid-span deflection from the interpolation : $v(0) = H_2(0)\theta_1 + H_4(0)\theta_2 = (L/8)(\theta_1 - \theta_2) = -qL^4/(96EI) = -0.0104\ qL^4/EI$, whereas exact $= -5qL^4/(384EI) = -0.0130\ qL^4/EI$. The error ($\approx$ 20 %) is removed by using more elements.

Convergence check : cantilever tip load with 1, 2 elements

Two elements of 500 mm, $EI = 2\times10^{12}$ N · mm^2, P = 1000 N at the tip. Unknowns v_2, θ_2, v_3, θ_3 after assembling two 4×4 matrices overlapping at node 2 : v_2 = -0.0521 mm, θ_2 = -0.000188 rad, v_3 = -0.1667 mm, θ_3 = -0.000250 rad. These coincide with the exact values ($PL^3/3EI$ = 0.1667 mm), because Hermite cubics can represent the exact cubic deflection curve of a tip-loaded cantilever.

Shear force and moment from FE results

- Bending moment in an element : $M(\xi) = EI\,[B]\{q\}$; at node 1 $M_1 = EI/L^2$ $[-6, -2L, 6, -4L]\{q\}$, at node 2 $M_2 = EI/L^2$ $[6, 4L, -6, 2L]\{q\}$.
- Shear force $V = dM/dx = (EI/L^3)$ $[12\ ,\ 6L\ ,\ -12\ ,\ 6L]\{q\}$ (constant in the element since the displacement is cubic).
- Alternatively use the nodal equilibrium : element end forces $\{f\} = [k]\{q\} - \{\text{fixed-end loads}\}$.

Truss vs Bar vs Beam : comparison

Element	Nodes	dof / node	Shape fn	Matrix size	Continuity
1-D bar	2	1 (u)	linear	2 × 2	C^0
Plane truss	2	2 (u,v)	linear	4 × 4	C^0
Beam	2	2 (v,θ)	Hermite cubic	4 × 4	C^1

2.8 Problem 9 : cantilever with UDL, two elements

Cantilever $L = 1000$ mm, $EI = 2\times10^{12}$ N · mm^2, UDL $q = 10$ N/mm downward. Two elements, $L_e = 500$ mm. Element load vector (downward q) : $\{f\} = -qL_e/2\ [1,\ L_e/6,\ 1,\ -L_e/6]^T = [-2500,\ -208333,\ -2500,\ 208333]^T$ (N, N · mm).

Assemble : the shared node 2 receives the sum of the contributions - $F_{v2} = -5000$ N, $F_{\theta 2} = 0$ N · mm (end moments cancel), $F_{v3} = -2500$ N, $F_{\theta 3} = 208333$ N · mm.

Apply $v_1 = \theta_1 = 0$ and solve the 4×4 reduced system : $v_2 = -0.2214$ mm, $\theta_2 = -0.000729$ rad, $v_3 = -0.6250$ mm, $\theta_3 = -0.000833$ rad. Exact tip deflection $-qL^4/(8EI) = -0.6250$ mm : nodal values are exact for a uniformly loaded beam.

Check: Nodal values for beams are exact for any load that is a polynomial of degree ≤ 3 when consistent nodal loads are used; the error lies only in the interpolated curve inside the element. Check the sum of vertical nodal loads = total load (here 10×1000 = 10 000 N).

Problem 10 : numerical element matrix and stress of one inclined bar

Bar from node 1 (0,0) to node 2 (300,400) mm; $A = 500\ mm^2$, $E = 2\times10^5\ N/mm^2$. $L = 500\ mm$, $l = 0.6$, $m = 0.8$, $AE/L = 2\times10^5\ N/mm$.

$$[k] = \begin{bmatrix} 72000 & 96000 & -72000 & -96000 \\ 96000 & 128000 & -96000 & -128000 \\ -72000 & -96000 & 72000 & 96000 \\ -96000 & -128000 & 96000 & 128000 \end{bmatrix}$$

Node 1 fixed, node 2 displaced $u_2 = 0.10\ mm$, $v_2 = 0.05\ mm$. Elongation of the bar $= l\ u_2 + m\ v_2 = 0.06 + 0.04 = 0.10\ mm$. Strain $= 0.10/500 = 2\times10^{-4}$, so $\sigma = E\varepsilon = 40\ N/mm^2$ (tension) and axial force $= 40\times500 = 20\ 000\ N$. Forces at node 2 : $F_x = 12000\ N$, $F_y = 16000\ N$ ($= 20\ 000 \times l$ and $\times\ m$).

Common mistakes to avoid

- Using degrees/radians wrongly in l, m; mixing units (N, mm, MPa); forgetting that dof of node i are (2i−1, 2i).
- Applying the support condition of a **roller** wrongly (only the dof normal to the rolling surface is zero).
- In beams : writing the load vector with wrong signs (check direction of q and sign of end moments).

Formula sheet (Unit II)

Item	Truss element	Beam element
dof / element	4 (u,v at 2 nodes)	4 (v,θ at 2 nodes)
Stiffness	AE/L [l^2, lm, ...]	EI/L^3 [12, 6L, ...]
Strain / curvature	$\varepsilon = (1/L)[-l\ -m\ l\ m]\{q\}$	$\kappa = (1/L^2)[6\xi\ ...]\{q\}$
Stress	$\sigma = E\,\varepsilon$	$\sigma = -E\,y\,\kappa$ or $M\,y\,/\,I$
Load vector	nodal loads only	$qL/2$ [1, L/6, 1, −L/6]

Short answer questions

Q1. Why is a truss element different from a bar element ? - Members are oriented at an angle; each node has 2 dof so a transformation from local to global axes is needed.

Q2. What is the size of the truss element matrix ? - 4×4 (2 nodes × 2 dof) in 2-D; 6×6 in 3-D.

Q3. Why Hermite polynomials for beams ? - Slope continuity (C^1) is necessary because strain energy contains d^2v/dx^2.

Q4. What is the degree of the beam interpolation ? - Cubic (4 dof per element).

Q5. What are consistent nodal loads ? - Loads derived from the same shape functions, $\{f\} = \int [H]^T q \, dx$; equal to fixed-end reactions.

Q6. Sign of stress in truss member ? - positive ⇒ tension, negative ⇒ compression.

Q7. Number of rigid-body modes of a beam element ? - Two (translation and rotation); a plane truss element has three (2 translations + 1 rotation) in the global system.

Expected questions

- Derive the stiffness matrix of a 2-D truss element; write stress expression. Solve a 3-member truss for displacements, stresses and reactions.
- Derive Hermite shape functions for a beam; derive [k] and load vector for UDL. Solve cantilever / simply supported beam using 1-2 elements.
- Why is C^1 continuity required for beam elements but C^0 suffices for bars?

UNIT III : Two-Dimensional Analysis

3.1 Introduction

Two-dimensional problems are solved when the body can be idealised in one plane : **plane stress** (thin plate in its own plane, $\sigma_z = 0$), **plane strain** (long prismatic body, $\varepsilon_z = 0$) or **axisymmetric** solids. Each node has **2 dof** (u, v). The simplest 2-D element is the **3-noded triangle** - the **Constant Strain Triangle (CST)**.

- Triangles fit curved and irregular boundaries easily and are used for automatic meshing.
- Nodes are numbered **anticlockwise** inside each element so that the area A is positive.
- Element dof vector $\{q\} = [q_1\ q_2\ q_3\ q_4\ q_5\ q_6]^T = [u_1\ v_1\ u_2\ v_2\ u_3\ v_3]^T$.

CST: 3 nodes, 6 dof
$\{q\} = [q_1 \dots q_6]^T$
ε = const over element

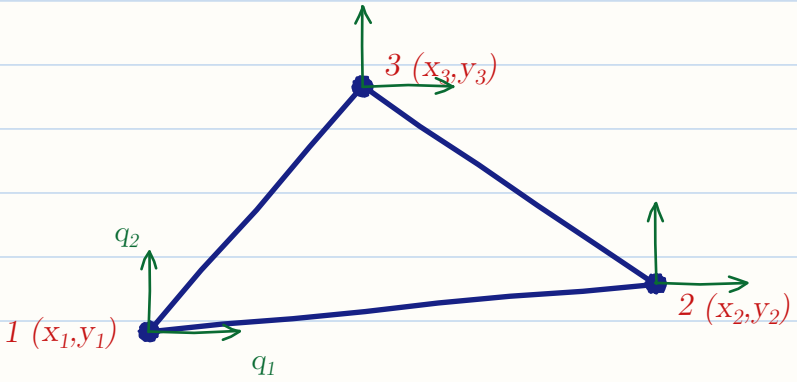


Fig 3.1 Constant strain triangle

3.2 Displacement field and shape functions

Three nodes per variable ⇒ 3 constants, a complete linear polynomial : $u = \alpha_1 + \alpha_2 x + \alpha_3 y$, $v = \alpha_4 + \alpha_5 x + \alpha_6 y$. Solving with nodal values :

$$u = N_1u_1 + N_2u_2 + N_3u_3,\ v = N_1v_1 + N_2v_2 + N_3v_3$$

$$N_i = (a_i + b_i x + c_i y)/2A \quad (i = 1,2,3)$$

with (cyclic order 1→2→3) : $a_1 = x_2y_3 - x_3y_2$, $b_1 = y_2 - y_3$, $c_1 = x_3 - x_2$ (and similarly for 2, 3 by cycling the subscripts), and

$$2A = x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) = \det [1\ x_1\ y_1\ ;\ 1\ x_2\ y_2\ ;\ 1\ x_3\ y_3]$$

Properties : $N_i = 1$ at node i, 0 at the others ; $N_1+N_2+N_3 = 1$; N_i varies linearly and is identical to the **area co-ordinate** L_i = (area of sub-triangle opposite node i)/A. Along an edge the displacement is linear, hence the displacement is continuous between adjacent elements (compatible).

$$\{u,v\}^T = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 \end{bmatrix}$$

3.3 Strain-displacement matrix and stiffness matrix

$\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$. Since $\partial N_i/\partial x = b_i/2A$ and $\partial N_i/\partial y = c_i/2A$:

$$[B] = 1/(2A) \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

Key point: *[B] contains only nodal coordinates - it is* **constant** *over the element. Hence strains and stresses are constant ⇒ the name CST. The stress is best interpreted at the centroid; fine meshes are needed where stress gradients are high.*

$$\{\varepsilon\} = [B]\{q\} \quad \{\sigma\} = [D][B]\{q\}$$

$$[k] = \int [B]^T[D][B]\, t\, dA = t\, A\, [B]^T[D][B] \; (6 \times 6, \text{symmetric})$$

[D] is the plane-stress or plane-strain matrix of Unit I; t = thickness (for plane strain take t = 1 per unit length).

Load vector

- ***Body force*** *(X, Y per unit volume, e.g. weight $Y = -\rho g$) : $\{f_b\} = (tA/3)\,[X\ Y\ X\ Y\ X\ Y]^T$ (one third of the total force at each node).*
- ***Traction (distributed edge load)*** *T_x, T_y per unit area on edge i-j of length L_{ij} : $\{f_T\} = (t\, L_{ij}/2)\,[T_x\ T_y\ T_x\ T_y]^T$ on the two edge nodes (zero at the third node). For a linearly varying load integrate with N_i.*
- ***Point loads*** *are applied directly at the nodes (always place a node under a concentrated load).*
- ***Initial strain / thermal*** *: $\{f_{th}\} = t\, A\, [B]^T[D]\{\varepsilon_0\}$, $\{\varepsilon_0\} = \alpha\Delta T\,[1\ 1\ 0]^T$ (plane stress).*

Treatment of boundary conditions

Global equation $[K]\{Q\} = \{F\}$ with size $2 \times$ (number of nodes). **Elimination** : delete rows and columns of restrained dof ($u = 0$ or $v = 0$). **Penalty** : add large C to the diagonal term. For an **inclined roller** support, use a multi-point constraint (transformation) or penalty : $\beta_1 Q_i + \beta_2 Q_j = \beta_0$. Reactions : $\{R\} = [K]\{Q\} - \{F\}$ for the restrained rows.

Stress calculation

$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{[\ ((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^{\ 2}\ ]},\ \tan 2\varphi = 2\tau_{xy}/(\sigma_x-\sigma_y)$$

$$\text{Von Mises} : \sigma_{vm} = \sqrt{(\ \sigma_x^{\ 2} - \sigma_x\sigma_y + \sigma_y^{\ 2} + 3\tau_{xy}^{\ 2}\ )}\ \text{(plane stress)}$$

3.4 Solved Problem 1 : stresses in one triangle

Triangle 1 (0,0), 2 (40,0), 3 (0,30) mm, $t = 5$ mm, $E = 2\times10^5$ N/mm^2, $\nu = 0.3$, plane stress. Nodal displacements $u_2 = 0.02$ mm, $v_3 = 0.015$ mm, all others zero. Find strains and stresses.

$A = ½\times40\times30 = 600$ mm^2. $b_1 = -30$, $b_2 = 30$, $b_3 = 0$; $c_1 = -40$, $c_2 = 0$, $c_3 = 40$.

$$[B] = 1/1200 \begin{bmatrix} -30 & 0 & 30 & 0 & 0 & 0 \\ 0 & -40 & 0 & 0 & 0 & 40 \\ -40 & -30 & 0 & 30 & 40 & 0 \end{bmatrix}$$

Strains $\{\varepsilon\} = [B]\{q\}$: $\varepsilon_x = b_2u_2/2A = 30\times0.02/1200 = 0.00050$, $\varepsilon_y = c_3v_3/2A = 40\times0.015/1200 = 0.00050$, $\gamma_{xy} = (c_2u_2 + b_3v_3 + \ldots)/2A = 0.00000$.

Stresses $\{\sigma\} = [D]\{\varepsilon\}$ with $[D] = (E/(1-\nu^2))\ [1\ \nu\ 0\ ;\ \nu\ 1\ 0\ ;\ 0\ 0\ (1-\nu)/2]$: $\sigma_x = 142.86$ N/mm^2, $\sigma_y = 142.86$ N/mm^2, $\tau_{xy} = 0.00$ N/mm^2.

Principal stresses $\sigma_1 = 142.86$, $\sigma_2 = 142.86$ N/mm^2; von Mises $= 142.86$ N/mm^2.

3.5 Solved Problem 2 : plate with two CST elements

Rectangular plate 40 × 30 mm, $t = 5$ mm, $E = 2\times10^5$ N/mm^2, $\nu = 0.3$ (plane stress). Left edge (nodes 1, 4) fixed; vertical load $P = 1000$ N downward at node 3. Elements : (1) nodes 1-2-3, (2) nodes 1-3-4 (both anticlockwise).

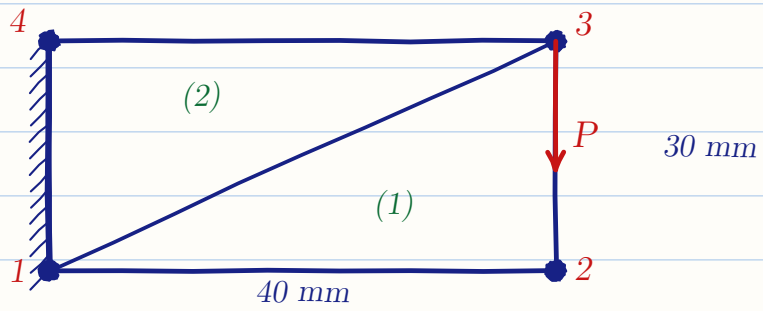


Fig 3.2

Element 1 : (0,0), (40,0), (40,30); ***Element 2*** : (0,0), (40,30), (0,30). Both have area $A = 600$ mm^2.

Element 1 : b = [-30, 30, 0], c = [0, -40, 40]. Element 2 : b = [0, 30, -30], c = [-40, 0, 40].

$[k] = tA[B]^T[D][B]$ for each element is a 6×6 matrix (D for plane stress : $E/(1-\nu^2)$ $= 2.198\times10^5$). Assembling (element 1 dof = 1,2,3,4,5,6 ; element 2 dof = 1,2,5,6,7,8) gives the 8×8 [K]. Boundary conditions $u_1 = v_1 = u_4 = v_4 = 0$ leave Q_3, Q_4, Q_5, Q_6 :

$$10^5 \times \begin{bmatrix} 6.68 & -3.57 & -2.56 & 1.65 \\ -3.57 & 8.77 & 1.92 & -7.33 \\ -2.56 & 1.92 & 6.68 & 0 \\ 1.65 & -7.33 & 0 & 8.77 \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = [0,\ 0,\ 0,\ -1000]^T$$

(entries shown in units of 10^5 N/mm). Solution :

u_2	v_2	u_3	v_3
-0.00090 mm	-0.00461 mm	0.00098 mm	-0.00483 mm

Stresses (N/mm^2) from $\{\sigma\} = [D][B]\{q\}$:

Element	σ_x	σ_y	τ_{xy}	σ_1	σ_2	σ_{vm}
1	-5.40	-3.04	-4.05	0.00	-8.44	8.44
2	5.40	1.62	-9.28	12.98	-5.96	16.78

Reactions at the fixed edge : R_{1x} = 1333.3, R_{1y} = 141.8, R_{4x} = -1333.3, R_{4y} = 858.2 N. ΣF_y = 1000.0 N (balances P = 1000 N) - check.

Remark: With only two CST elements the solution is stiff and inaccurate in bending (beam theory gives a much larger tip deflection). Refine the mesh : the CST converges slowly for bending-dominated problems. Stress at the support is not captured; compare with $\sigma = My/I$ at the root.

3.6 Axisymmetric Solids

A solid of revolution loaded symmetrically about its axis (cylinders, pressure vessels, pipes, flywheels, rotors, nozzles, pistons) is analysed on a ***single r-z cross-section***. The geometry, material, loads and BC do not vary with θ, so the displacement has two components : u (radial) and w (axial), $v_\theta = 0$.

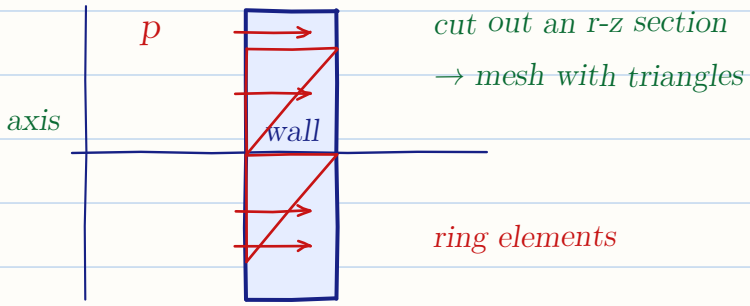


Fig 3.3 Axisymmetric solid → ring elements

The 2-D element is a triangle in the r-z plane, which, when revolved through 360°, becomes a ***ring*** (torus of triangular section) joined to others along nodal circles. Nodal dof : (u_i, w_i) with $\{q\} = [u_1\ w_1\ u_2\ w_2\ u_3\ w_3]^T$.

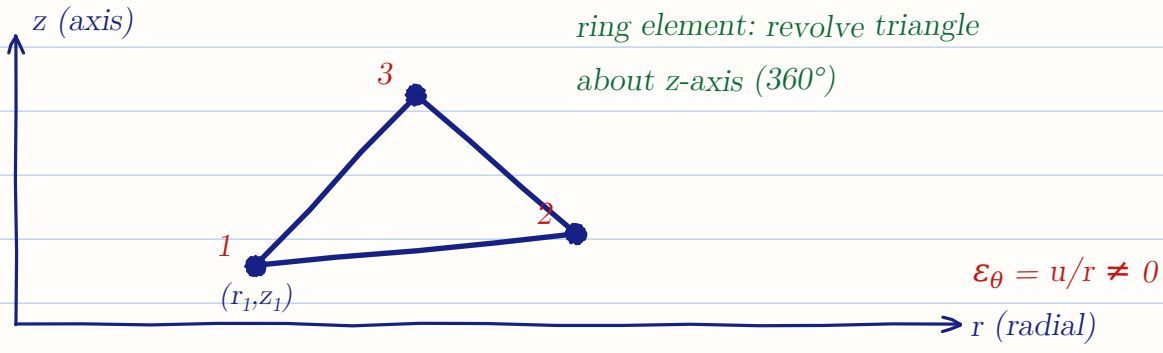


Fig 3.4 Triangular ring element in the r-z plane

Strain-displacement relations

$$\{\varepsilon\} = [\varepsilon_r \; \varepsilon_z \; \varepsilon_\theta \; \gamma_{rz}]^T = \begin{bmatrix} \partial u/\partial r \\ \partial w/\partial z \\ u/r \\ \partial u/\partial z + \partial w/\partial r \end{bmatrix}$$

The hoop strain $\varepsilon_\theta = u/r$ is the new feature : a radial displacement stretches the circumference. Therefore there are **4 stress/strain components** - σ_r, σ_z, σ_θ, τ_{rz}.

Stress-strain matrix

$$[D] = E(1-\nu)/((1+\nu)(1-2\nu)) \begin{bmatrix} 1 & \nu/(1-\nu) & \nu/(1-\nu) & 0 \\ \nu/(1-\nu) & 1 & \nu/(1-\nu) & 0 \\ \nu/(1-\nu) & \nu/(1-\nu) & 1 & 0 \\ 0 & 0 & 0 & (1-2\nu)/(2(1-\nu)) \end{bmatrix}$$

Shape functions and [B] matrix

Same as the CST with (x, y) → (r, z) : $N_i = (\alpha_i + \beta_i r + \gamma_i z)/2A$, $\alpha_1 = r_2 z_3 - r_3 z_2$, $\beta_1 = z_2 - z_3$, $\gamma_1 = r_3 - r_2$ (cyclic).

$$[B] = 1/(2A) \begin{bmatrix} \beta_1 & 0 & \beta_2 & 0 & \beta_3 & 0 \\ 0 & \gamma_1 & 0 & \gamma_2 & 0 & \gamma_3 \\ f_1 & 0 & f_2 & 0 & f_3 & 0 \\ \gamma_1 & \beta_1 & \gamma_2 & \beta_2 & \gamma_3 & \beta_3 \end{bmatrix}$$

where $f_i = \alpha_i/r + \beta_i + \gamma_i z/r$. Because f_i varies with (r, z), [B] is not constant; it is evaluated at the **centroid** $\bar{r} = (r_1+r_2+r_3)/3$, $\bar{z} = (z_1+z_2+z_3)/3$ (good approximation for fine mesh). Hence the element stresses are taken constant and :

$$[k] = 2\pi \, \bar{r} \, A \, [B]^T[D][B] \; (6 \times 6)$$

Remember: Volume = $2\pi \, \bar{r} \, A$ (Pappus theorem). There is no thickness t; the $2\pi \, \bar{r}$ factor replaces it. The element near the axis ($r \to 0$) needs care : the term u/r is singular; use $\bar{r}$ at the centroid, fix $u = 0$ on the axis.

Load vectors

- **Body force** (e.g. centrifugal force $\rho\omega^2 r$ in radial direction, weight along z) : $\{f\} = (2\pi\ \bar{r}\ A\ /3)\ [F_r\ F_z\ F_r\ F_z\ F_r\ F_z]^T$ with $F_r = \rho\omega^2\bar{r}$.
- **Surface pressure** p on an edge i-j of length L_e (e.g. internal pressure on the bore) : $\{f_i\} = (\pi\ p\ L_e/3)(2r_i + r_j)$, $\{f_j\} = (\pi\ p\ L_e/3)(r_i + 2r_j)$, applied in the direction normal to the surface (radial for the bore). Simplified : $f = 2\pi\ \bar{r}_{edge}\ p\ L_e/2$ at each node.
- **Ring / line loads** : a load P per unit circumference at radius r gives a nodal force $2\pi\ r\ P$.

3.7 Solved Problem 3 : axisymmetric triangle

Triangle with nodes 1 (r=20, z=0), 2 (30, 0), 3 (20, 10) mm. $E = 2\times10^5$ N/mm^2, $\nu = 0.3$. Find [B], [D] and [k].

$A = ½\times10\times10 = 50$ mm^2; $\bar{r} = (20+30+20)/3 = 23.33$ mm, $\bar{z} = 3.33$ mm. α = [300, -200, 0], β = [-10, 10, 0], γ = [-10, 0, 10].

$[D] = 2.692\times10^5 \times$ [1 0.4286 0.4286 0 ; 0.4286 1 0.4286 0 ; 0.4286 0.4286 1 0 ; 0 0 0 0.2857] , with $E(1-\nu)/((1+\nu)(1-2\nu)) = 2.692\times10^5$ N/mm^2.

[B] (evaluated at the centroid), entries $\times\ 10^{-2}$:

$$[B] = \begin{bmatrix} -10.000 & 0.000 & 10.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & -10.000 & 0.000 & 0.000 & 0.000 & 10.000 \\ 1.429 & 0.000 & 1.429 & 0.000 & 1.429 & 0.000 \\ -10.000 & -10.000 & 0.000 & 10.000 & 10.000 & 0.000 \end{bmatrix}$$

$[k] = 2\pi\ \bar{r}\ A\ [B]^T[D][B]$ with $2\pi\ \bar{r}\ A = 2\pi\times23.33\times50 = 7330$ mm^3. Result in units of 10^6 N/mm :

$$[k] = \begin{bmatrix} 23.361 & 12.889 & -19.333 & -5.639 & -6.444 & -7.250 \\ 12.889 & 25.374 & -9.666 & -5.639 & -6.847 & -19.736 \\ -19.333 & -9.666 & 22.555 & 0.000 & 1.611 & 9.666 \\ -5.639 & -5.639 & 0.000 & 5.639 & 5.639 & 0.000 \\ -6.444 & -6.847 & 1.611 & 5.639 & 6.042 & 1.208 \\ -7.250 & -19.736 & 9.666 & 0.000 & 1.208 & 19.736 \end{bmatrix}$$

Check : symmetric and singular - the sum of the radial-dof rows is not zero because rigid-body radial translation is not a rigid-body mode of a ring (it stretches the hoop).

Comparison : plane stress / plane strain / axisymmetric

Item	Plane stress / strain	Axisymmetric
Coordinates	x, y	r, z
dof / node	u, v	u, w
Strains	3	4 (adds $\varepsilon_\theta = u/r$)
[B] matrix	constant	varies - use centroid
[k]	$t\ A\ B^T D B$	$2\pi\ \bar{r}\ A\ B^T D B$
Typical use	plates, dams	vessels, rotors, pipes

3.8 More on CST and Axisymmetric Elements

Derivation of the CST shape functions

1. Write $u = \alpha_1 + \alpha_2 x + \alpha_3 y$ and apply $u = u_1, u_2, u_3$ at the three nodes : $\{u_1\ u_2\ u_3\}^T = [\ 1\ x_1\ y_1\ ;\ 1\ x_2\ y_2\ ;\ 1\ x_3\ y_3\ ]\{\alpha_1\ \alpha_2\ \alpha_3\}^T$.
2. Invert the 3×3 matrix; its determinant equals $2A$. The inverse $= (1/2A)\ [\ a_1\ a_2\ a_3\ ;\ b_1\ b_2\ b_3\ ;\ c_1\ c_2\ c_3\ ]$ (co-factor matrix).
3. Substitute $\{\alpha\}$ back : $u = [1\ x\ y](1/2A)[...]\{u_i\} \Rightarrow N_i = (a_i + b_i x + c_i y)/2A$. Same for v.
4. Check : at node 1, $N_1 = (a_1 + b_1 x_1 + c_1 y_1)/2A = 2A/2A = 1$.

Integration with area co-ordinates

$$\int_A L_1^{\,a}\, L_2^{\,b}\, L_3^{\,c}\, dA = a!\ b!\ c!\ 2A\ /\ (a + b + c + 2)!$$

Examples : $\int L_i\, dA = A/3$ (body force $\Rightarrow$ one-third at each node) ; along an edge $\int L_i\, ds = L_e/2$; $\int L_i L_j\, dA = A/12$ $(i \neq j)$, $\int L_i^2\, dA = A/6$. Used for load vectors and mass matrices.

Problem 4 : load vectors for a triangle

Triangle (0,0), (40,0), (0,30), $t = 5$ mm. (a) **Self weight** : steel $\rho g = 7.85\times10^{-5}$ N/mm^3 acting in $-y$. $tA = 5\times600 = 3000$ mm^3, so total weight $W = 0.2355$ N. Nodal load $f_y = -tA\,\rho g/3 = -0.0785$ N at each node (x-components zero).

(b) **Pressure** $p = 10$ N/mm^2 acting normal to the inclined edge 2-3 (length 50 mm), pointing inwards. Direction cosines of the outward normal : (0.6, 0.8) $\Rightarrow$ $T_x = -6$, $T_y = -8$ N/mm^2. $f = tL_e/2\,[T_x, T_y] = (5\times50/2) = 125 \times [-6, -8] = [-750, -1000]$ N at each of nodes 2 and 3; node 1 gets zero.

$$\{f_T\} = \begin{bmatrix} 0 \\ 0 \\ -750 \\ -1000 \\ -750 \\ -1000 \end{bmatrix}$$

Check : resultant $= \sqrt{(1500^2 + 2000^2)} = 2500$ N $= p\,t\,L = 10 \times 5 \times 50$ OK (acting on the edge, normal to it).

Problem 5 : plane strain [D] and the effect of ν

For $E = 2\times10^5$ N/mm^2, $\nu = 0.3$:

$$\text{Plane stress } [D]\ (\times10^5) = \begin{bmatrix} 2.198 & 0.659 & 0.000 \\ 0.659 & 2.198 & 0.000 \\ 0.000 & 0.000 & 0.769 \end{bmatrix}$$

$$\text{Plane strain } [D]\ (\times10^5) = \begin{bmatrix} 2.692 & 1.154 & 0.000 \\ 1.154 & 2.692 & 0.000 \\ 0.000 & 0.000 & 0.769 \end{bmatrix}$$

Plane strain is stiffer ($D_{11} = 2.692\times10^5$ against 2.198×10^5). In the plane-strain case $\sigma_z = \nu(\sigma_x + \sigma_y) \neq 0$ must be added when finding von Mises stress.

Problem 6 : thermal load on a CST

Plane stress triangle, $t = 5$, $A = 600\ mm^2$, $\alpha = 12\times10^{-6}/°C$, free expansion $\Delta T = 100\ °C$ produces initial strain $\varepsilon_0 = [1.2\times10^{-3},\ 1.2\times10^{-3},\ 0]^T$. Thermal load $\{f_{th}\} = tA[B]^T[D]\{\varepsilon_0\}$. $[D]\{\varepsilon_0\} = (E/(1-\nu))\ \alpha\Delta T\ [1\ 1\ 0]^T = 2.857\times10^5\times1.2\times10^{-3} = 342.9\ N/mm^2$ in both σ_x, σ_y. Then $f = tA/(2A)\ [b_1, c_1, b_2, c_2, b_3, c_3]\times342.9 = (t/2)(342.9)[b_1\ c_1\ b_2\ c_2\ b_3\ c_3]^T$ (b, c in mm). If the element is completely free, the displacements absorb the load and $\sigma = 0$.

Problem 7 : axisymmetric loads on a thick cylinder

Thick cylinder, mean section : bore pressure $p = 10\ N/mm^2$ acting on the inner surface $r = 50$ mm over an edge of length 20 mm (nodes i, j both at $r = 50$) :

$$f_i = f_j = \pi\ p\ L_e(2r_i + r_j)/3 = \pi\times10\times20\times150/3 = 31416\ N\ (radial)$$

Check with $2\pi\ r\ p\ L_e/2 = \pi\times50\times10\times20 = 31416\ N$ OK. For a spinning disc ($\rho = 7.85\times10^{-9}\ N\cdot s^2/mm^4$, $\omega = 300\ rad/s$) the body force at the element of Problem 3 is $F_r = \rho\omega^2\bar{r} = 7.85\times10^{-9}\times9\times10^4\times23.33 = 0.0165\ N/mm^3$; total radial force = $F_r \times 2\pi\ \bar{r}\ A = 120.8\ N$, i.e. $40.3\ N$ at each node.

Higher-order triangle (LST) in brief

- 6-noded triangle (3 corner + 3 mid-side nodes) : complete quadratic $u = \alpha_1 + \alpha_2x + \alpha_3y + \alpha_4x^2 + \alpha_5xy + \alpha_6y^2$, 12 dof.
- Strain varies **linearly** (Linear Strain Triangle) so it models bending and stress concentrations much better than the CST; shape functions in area co-ordinates : $N_1 = L_1(2L_1 - 1)$, $N_4 = 4L_1L_2$, ...

Short answer questions

Q1 Why is CST called constant strain triangle ? - $[B]$ contains only nodal coordinates, so ε and σ are constant over the element.

Q2 Size of $[k]$ for CST ? - 6×6 ; for a 4-noded quadrilateral 8×8.

Q3 Why must nodes be numbered anticlockwise ? - To obtain positive area A ($2A$ is the determinant).

Q4 Why does $[B]$ of the axisymmetric element depend on r, z ? - Hoop strain $\varepsilon_\theta = u/r$.

Q5 What is the volume element of an axisymmetric solid ? - $dV = 2\pi\ r\ dr\ dz$.

Q6 Where do we use plane strain ? - Dams, retaining walls, long pipes, tunnels (ε_z

$= 0$).

Problem 8 : stress in an axisymmetric ring element

For the triangle of Problem 3, let the nodal displacements be $u_1 = 0.004$, $w_1 = 0$, $u_2 = 0.006$, $w_2 = 0$, $u_3 = 0.004$, $w_3 = 0.002$ mm. Then $\{\varepsilon\} = [B]\{q\}$ gives

ε_r	ε_z	ε_θ	γ_{rz}
0.000200	0.000200	0.000200	0.000000

and $\{\sigma\} = [D]\{\varepsilon\}$: $\sigma_r = 100.0$, $\sigma_z = 100.0$, $\sigma_\theta = 100.0$, $\tau_{rz} = 0.0$ N/mm^2. Note that $\sigma_\theta \neq 0$ (hoop stress) even though no circumferential displacement exists.

Problem 9 : equivalent load of a linearly varying edge pressure

Pressure varies from p_i at node i to p_j at node j along an edge of length L_e (thickness t). Using the linear shape functions along the edge :

$$f_i = t\,L_e(2p_i + p_j)/6 \quad f_j = t\,L_e(p_i + 2p_j)/6$$

Example : $t = 5$, $L_e = 30$, $p_i = 0$, $p_j = 12$ N/mm^2 : $f_i = 5\times30\times12/6 = 300$ N, $f_j = 5\times30\times24/6 = 600$ N; total 900 N $= t\,L_e(p_i+p_j)/2$. Always check that the total of nodal loads equals the resultant of the distributed load.

Complete procedure for a 2-D (plane) problem

1. Idealise : decide plane stress / plane strain / axisymmetric; use symmetry to cut the model.
2. Mesh : number nodes and elements (anticlockwise); tabulate connectivity and coordinates.
3. For each element compute b_i, c_i, A, $[B]$, $[D]$, $[k] = tA\,B^TDB$.
4. Element load vectors (body force, traction, thermal) and point loads.
5. Assemble $[K]$ (dof of node $i = 2i{-}1,\ 2i$) and $\{F\}$.
6. Apply BC, solve for $\{Q\}$.
7. Recover $\{\varepsilon\} = [B]\{q\}$, $\{\sigma\} = [D]\{\varepsilon\}$, principal and von Mises stress, reactions.

Element comparison

Element	Nodes	dof	Polynomial	Strain
CST	3	6	linear	constant
LST	6	12	quadratic	linear
Q4	4	8	bilinear (1,x,y,xy)	linear in part
Q8	8	16	serendipity quad.	quadratic
Axisym. tri.	3	6	linear in r,z	varies (use centroid)

Tips: Boundary-condition tips : symmetry plane ⇒ normal displacement = 0 ; antisymmetry ⇒ tangential displacement = 0 ; for axisymmetric solids, nodes on the axis have $u = 0$. Never leave a rigid-body mode (3 in plane, 4 in 3-D axisymmetric with axial translation).

Mesh guidelines and expected questions

- Use fine mesh where stress gradients are high (holes, notches, fillets, load points); avoid very thin triangles (aspect ratio < 3). Make node on every point load, material change, support.
- Derive shape functions, [B] and [k] of a CST; explain plane stress vs plane strain with [D]. Solve numerical problems (1-2 elements) for stresses, load vector.
- Derive [B], [D] and [k] for a triangular axisymmetric element; why is [B] evaluated at the centroid ? Give load vector for pressure and centrifugal force.

UNIT IV : Quadrilateral Elements & Thermal Analysis

4.1 Isoparametric, Sub-parametric and Super-parametric elements

A general quadrilateral with straight edges cannot be described by a simple polynomial in x, y. The trick is to work in a **parent (natural) element** — a square $-1 \le \xi, \eta \le 1$ — and **map** it to the real element. Two sets of interpolation are used :

$$x = \Sigma N_i^g(\xi,\eta)\, x_i,\ y = \Sigma N_i^g\, y_i \text{ (geometry)}\quad u = \Sigma N_i^u(\xi,\eta)\, u_i \text{ (displacement)}$$

Type	Geometry nodes vs displacement nodes	Result
Isoparametric	same shape functions for both ($N^g = N^u$)	most widely used
Sub-parametric	geometry order < displacement order	curved geometry not needed, accurate u
Super-parametric	geometry order > displacement order	rarely used, may fail patch test

"Iso" = same, "sub" = fewer nodes for geometry than for the field, "super" = more nodes for geometry than for the field. Example : a 8-noded element with a straight-edge geometry described by 4 corner nodes only = **sub-parametric**; a 4-noded displacement field with 8-noded curved geometry = **super-parametric**.

Why use them?: Isoparametric elements pass the patch test and satisfy completeness / compatibility if the mapping is one-to-one (det $J > 0$ everywhere, element convex, interior angles < 180°).

4.2 Four-noded Quadrilateral (Q4)

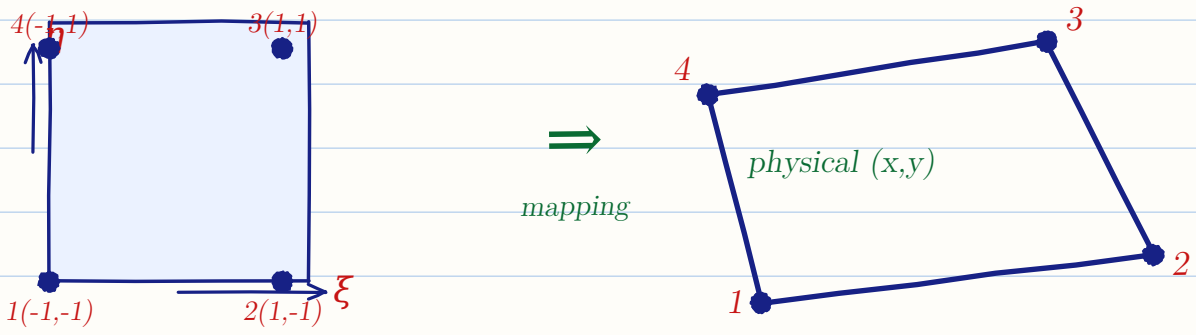


Fig 4.1 Q4 : parent element (ξ,η) and physical element (x,y)

Shape functions (bilinear; $N_i = 1$ at node i, 0 at other nodes) with node numbering counter-clockwise 1(−1,−1), 2(1,−1), 3(1,1), 4(−1,1) :

$$N_1 = (1-\xi)(1-\eta)/4\quad N_2 = (1+\xi)(1-\eta)/4\quad N_3 = (1+\xi)(1+\eta)/4\quad N_4 = (1-\xi)(1+\eta)/4$$

Properties : $\Sigma N_i = 1$; each N_i is linear along every edge ⇒ displacement compatible between neighbours. Displacement field has 8 dof : $\{q\} = [u_1\ v_1\ \ldots\ u_4\ v_4]^T$, contains terms 1, ξ, η, $\xi\eta$.

$$\{u,v\}^T = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix}$$

Jacobian, [B] and [k]

Derivatives of N_i are known in (ξ,η) but strains need (x,y). Chain rule :

$$\{\partial/\partial\xi\ ;\ \partial/\partial\eta\} = [J]\{\partial/\partial x\ ;\ \partial/\partial y\}, \quad [J] = \begin{bmatrix} \partial x/\partial\xi & \partial y/\partial\xi \\ \partial x/\partial\eta & \partial y/\partial\eta \end{bmatrix}$$

$$[J] = [dN/d\xi\ ;\ dN/d\eta]\ [x_1\ y_1\ ;\ \ldots\ ;\ x_4\ y_4]\ \{\partial N/\partial x\ ;\ \partial N/\partial y\} = [J]^{-1}\ \{\partial N/\partial\xi\ ;\ \partial N/\partial\eta\}$$

$$[B] = \begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} \end{bmatrix}$$

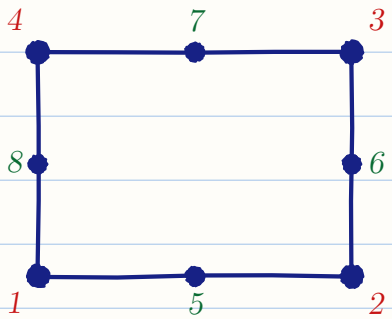
$$[k] = t \int_{-1}^{1}\int_{-1}^{1} [B]^T[D][B]\ det[J]\ d\xi\ d\eta$$

$dx\ dy = det[J]\ d\xi\ d\eta$. The integrand is a rational function of ξ, η (because of J^{-1}) for a general quad, so the integral is evaluated **numerically** (Gauss quadrature). For a rectangle J is constant and the integrand is polynomial.

Load vectors

- Body force : $\{f_b\} = t \iint [N]^T\{X\ Y\}^T\ det\ J\ d\xi\ d\eta$ (for uniform force on a rectangle, one quarter of the total at each node).
- Edge traction on side with nodes i-j : $\{f_T\} = t \int [N]^T\{T_x\ T_y\}^T\ ds$, $ds = (L_{ij}/2)\ d\xi$ ⇒ uniform traction gives $T\ t\ L_{ij}/2$ at each of the two nodes.

4.3 Eight-noded Quadrilateral (Q8, serendipity)



corner nodes 1-4

mid-side nodes 5-8

16 dof, curved edges

Fig 4.2 Q8 : corner nodes 1-4, mid-side nodes 5-8

Q8 has curved edges (parabolic) and 16 dof; polynomial contains 1, ξ, η, ξ^2, $\xi\eta$, η^2, $\xi^2\eta$, $\xi\eta^2$. Shape functions (serendipity family) :

Corner nodes *($i = 1..4$, ξ_i, $\eta_i = \pm 1$) : $N_i = (1/4)(1 + \xi\xi_i)(1 + \eta\eta_i)(\xi\xi_i + \eta\eta_i - 1)$*

Mid-side nodes *: $N_5 = \frac{1}{2}(1-\xi^2)(1-\eta)$, $N_6 = \frac{1}{2}(1+\xi)(1-\eta^2)$, $N_7 = \frac{1}{2}(1-\xi^2)(1+\eta)$, $N_8 = \frac{1}{2}(1-\xi)(1-\eta^2)$.*

Check : at node 5 (0,−1) $N_5 = \frac{1}{2}(1)(2) = 1$; sum of all $N = 1$; each N_i varies quadratically on edges, so the strain varies linearly - much better than Q4 for bending and curved boundaries. Stiffness : same formulae with 16×16 [k], integrated with 3×3 Gauss points.

> ***Remark:*** *Q4 shows shear locking in pure bending (too stiff), Q8 is far better. Q9 (Lagrange, adds centre node) has shape functions N = products of 1-D quadratic Lagrange polynomials.*

4.4 Numerical Integration (Gauss Quadrature)

Gauss-Legendre rule : $\int_{-1}^{1} f(\xi)\, d\xi \approx \Sigma\, w_i\, f(\xi_i)$. With n points it integrates exactly a polynomial of degree ***2n − 1****.*

n	Gauss points ξ_i	Weights w_i	Exact up to degree
1	0	2	1
2	±0.57735	1 , 1	3
3	0 , ±0.77460	8/9 , 5/9	5
4	±0.33998 , ±0.86114	0.65215 , 0.34785	7

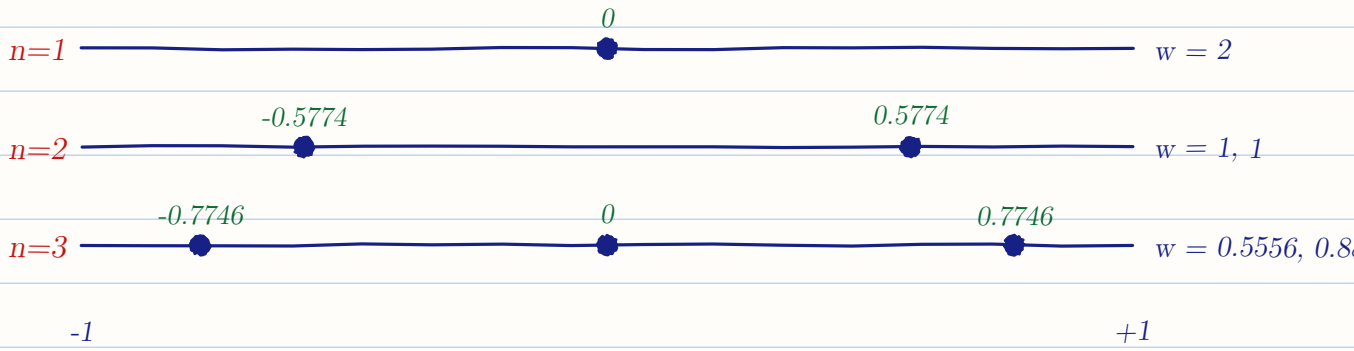


Fig 4.3 Gauss points on [−1, 1]

Two dimensions *: $\int\int f\, d\xi\, d\eta \approx \Sigma\Sigma\, w_i w_j\, f(\xi_i, \eta_j)$. For Q4 use 2×2 (4 points, w*

= 1, points (±0.57735, ±0.57735)); for Q8 use 3×3 (9 points).

General interval [a,b] : $x = (b-a)\xi/2 + (a+b)/2$, $dx = (b-a)/2\, d\xi$.

Example : Gauss integration

Evaluate $I = \int_{-1}^{1} (x^3 + 2x^2 + 1)\, dx$. Exact = 0 + 4/3 + 2 = 3.3333. Two-point rule : f(−0.57735) + f(0.57735) = 1.4742 + 1.8591 = 3.3333 (exact, since degree 3 ≤ 2n − 1 = 3).

One-point rule would give 2 f(0) = 2 (wrong, error 40 %). For $\int_1^3 x^2\, dx$: $x = \xi + 2$, $dx = d\xi \Rightarrow \int (\xi+2)^2 d\xi = 2.0 \cdot [f(-0.577)+f(0.577)]$ with $f = (\xi+2)^2$: 2.0204 + 6.6463 = 8.6667 = 26/3 (exact).

4.5 Solved Problems on the Quadrilateral

Problem 1 : mapping and Jacobian

Quadrilateral nodes 1 (0,0), 2 (4,0), 3 (5,3), 4 (1,3). (a) Find the global coordinates of the point $(\xi,\eta) = (0.5, 0.5)$; (b) Jacobian at that point.

$N = [0.0625, 0.1875, 0.5625, 0.1875]$: $N_1 = (0.5)(0.5)/4 = 0.0625$, $N_2 = (1.5)(0.5)/4 = 0.1875$, $N_3 = (1.5)(1.5)/4 = 0.5625$, $N_4 = (0.5)(1.5)/4 = 0.1875$.

$x = \Sigma N_i x_i = 0.1875\times4 + 0.5625\times5 + 0.1875\times1 = 3.7500$, $y = 0.5625\times3 + 0.1875\times3 = 2.2500$.

$$[J] = \begin{bmatrix} 2.000 & 0.000 \\ 0.500 & 1.500 \end{bmatrix}$$

$\det[J] = 2.000 \times 1.500 - 0.000 \times 0.500 = 3.000$ (> 0 ⇒ valid mapping). Check : det J > 0 at all four Gauss points as well; the element is convex.

Area of the quadrilateral = $\int\int \det J\, d\xi\, d\eta$ = 2×2 Gauss sum = 12.000 mm^2 (trapezium : ½(4+4)×3 = 12 OK.)

Problem 2 : stiffness matrix of a square Q4 element

Square of side a (any size), plane stress, thickness t, $\nu = 0.3$, 2×2 Gauss integration. det $J = a^2/4$ is constant, $J = (a/2)\, I$. The result is $k = E\, t \times$ (matrix below) :

$$[k]/(Et) = \begin{bmatrix} 0.495 & 0.179 & -0.302 & -0.014 & -0.247 & -0.179 & 0.055 & 0.014 \\ 0.179 & 0.495 & 0.014 & 0.055 & -0.179 & -0.247 & -0.014 & -0.302 \\ -0.302 & 0.014 & 0.495 & -0.179 & 0.055 & -0.014 & -0.247 & 0.179 \\ -0.014 & 0.055 & -0.179 & 0.495 & 0.014 & -0.302 & 0.179 & -0.247 \\ -0.247 & -0.179 & 0.055 & 0.014 & 0.495 & 0.179 & -0.302 & -0.014 \\ -0.179 & -0.247 & -0.014 & -0.302 & 0.179 & 0.495 & 0.014 & 0.055 \\ 0.055 & -0.014 & -0.247 & 0.179 & -0.302 & 0.014 & 0.495 & -0.179 \\ 0.014 & -0.302 & 0.179 & -0.247 & -0.014 & 0.055 & -0.179 & 0.495 \end{bmatrix}$$

Properties : symmetric ; each row sums to zero for the x-dof rows (rigid translation) ; diagonal terms positive ($k_{11} = 0.4945\ Et$). The matrix is independent of the size a for a square (stiffness $\propto t$) - a characteristic of 2-D elasticity.

4.6 Steady-state Heat Transfer

*Heat is transferred by **conduction** (Fourier : $q = -k\ dT/dx$), **convection** (Newton : $q = h(T_s - T_\infty)$) and radiation. In FEM the field variable is the temperature T, one dof per node. Governing equation (1-D, steady, with heat generation Q) :*

$$d/dx\ (\ k\ A\ dT/dx\ ) + Q = 0 \quad BC : T = T_0 \text{ (prescribed) or } -k\ dT/dx = h\ (T - T_\infty) \text{ (convection) or } q = q_0$$

Piece-wise linear temperature in a 2-noded element : $T = N_1T_1 + N_2T_2$, $N_1 = (1-\xi)/2$, $N_2 = (1+\xi)/2$. Applying the functional or Galerkin method :

$$[K_c] = kA/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \end{Bmatrix} = (QAL/2)[1,1]^T + \{q\}$$

Analogy with the bar : displacement ↔ temperature, force ↔ heat flow, AE/L ↔ kA/L. Hence the assembly and BC procedure are the same as Unit I.

- ***Convection at a node** (surface area A) : add hA to the diagonal term K_{ii} and $hA\ T_\infty$ to the load F_i.*
- ***Prescribed heat flux** q_0 at a node : add q_0A to F_i. **Insulated end** : nothing to add (natural BC).*
- *Heat flow through an element $Q = -kA\ (T_2 - T_1)/L$.*

Composite slab (wall)

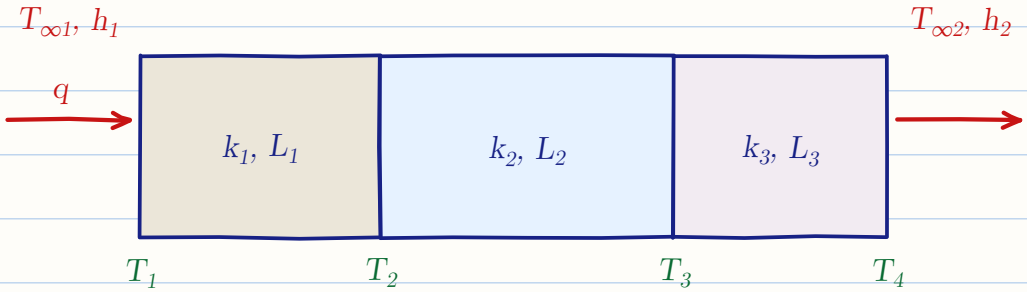


Fig 4.4 Three-layer composite wall

Each layer = one element of conductance k_iA/L_i; nodes at the interfaces. Both outer surfaces have convection (h_1, $T_{\infty 1}$ and h_2, $T_{\infty 2}$). Global equation for 3 layers (4 nodes) :

$$[K]\ (A=1)\ ;\ k_i = k_i/L_i \begin{bmatrix} k_1+h_1 & -k_1 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2+k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3+h_2 \end{bmatrix}$$

with load vector $F = [h_1T_{\infty 1},\ 0,\ 0,\ h_2T_{\infty 2}]^T$ (area $A = 1\ m^2$).

Problem 3 : numerical (3-layer wall)

Wall of layers $k = 20, 30, 50\ W/m \cdot K$, thickness 0.30, 0.15, 0.15 m. Inside gas $T_{\infty 1} = 800\ °C$, $h_1 = 25\ W/m^2K$; outside air $T_{\infty 2} = 20\ °C$, $h_2 = 15\ W/m^2K$. Area $= 1\ m^2$. Find interface temperatures and heat flux.

$k_1/L_1 = 20/0.3 = 66.67$, $k_2/L_2 = 30/0.15 = 200.00$, $k_3/L_3 = 50/0.15 = 333.33$ W/K.

$$\begin{bmatrix} 91.67 & -66.67 & 0.00 & 0.00 \\ -66.67 & 266.67 & -200.00 & 0.00 \\ 0.00 & -200.00 & 533.33 & -333.33 \\ 0.00 & 0.00 & -333.33 & 348.33 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{Bmatrix} = [20000,\ 0,\ 0,\ 300]^T$$

Solving : $T_1 = 559.38\ °C$, $T_2 = 469.15\ °C$, $T_3 = 439.07\ °C$, $T_4 = 421.03\ °C$. Heat flux $q = h_1(T_{\infty 1} - T_1) = 6015.4\ W/m^2$. Check with the resistance method : $q = (800 - 20)/(1/25 + 0.3/20 + 0.15/30 + 0.15/50 + 1/15) = 6015.4\ W/m^2$ OK.

4.7 Fin (Extended Surface) Analysis

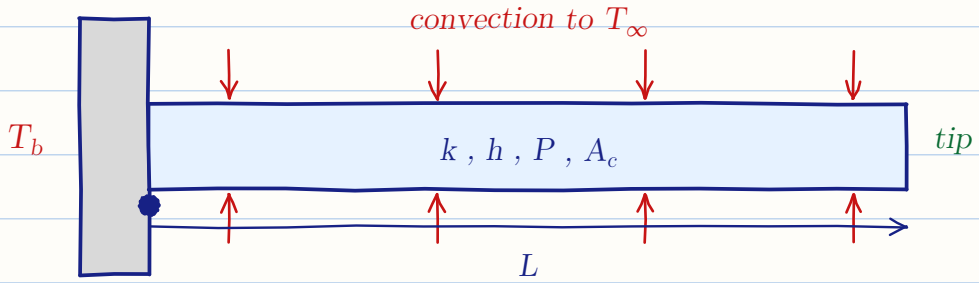


Fig 4.5 Straight fin attached to a hot wall

Fin governing equation (1-D, constant section, perimeter P, area A_c, convection h to ambient T_∞) :

$$k\,A_c\,d^2T/dx^2 - h\,P\,(T - T_\infty) = 0 \quad T(0) = T_b \text{ (base); tip : insulated or } -k\,dT/dx = h(T - T_\infty)$$

Element (2-noded) = conduction part + convection part :

$$[k_e] = kA_c/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + hPL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\{f_e\} = hPT_\infty L/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Tip convection (no insulation) : add hA_c to the last diagonal term and hA_cT_∞ to the last load term. Heat dissipated by the fin = heat conducted at the base : $Q = -kA_c(dT/dx)_{x=0}$; best evaluated from the ***reaction*** *at node 1 ($R_1 = [K]_{1,*}\{T\} - F_1$) because the element gradient is less accurate than the nodal values.*

Problem 4 : pin fin (insulated tip)

Aluminium-type pin fin : d = 10 mm, L = 100 mm, k = 60 W/m · K, h = 25 W/m^2K, T_b = 120 °C, T_∞ = 30 °C. In mm-W-K units : k = 0.06 W/mm · K, h = 25×10^{-6} W/mm^2K, $P = \pi d$ = 31.42 mm, A_c = 78.54 mm^2. Four elements, L_e = 25 mm.

kA_c/L_e = 0.1885 ; $hPL_e/6$ = 0.00327 ; $hPT_\infty L_e/2$ = 0.29452 W. The element matrix is $[k_e]$ = [0.1950, -0.1852 ; -0.1852, 0.1950] and the assembled system with T_1 = 120 °C gives :

x (mm)	0	25	50	75	100
T FEM (°C)	120.00	99.21	85.76	78.22	75.80
T exact (°C)	120.00	99.33	85.94	78.44	76.02

The FEM solution agrees with the analytical result $T = T_\infty + (T_b - T_\infty)\cosh m(L-x)/\cosh mL$, where $m = \sqrt{(hP/kA_c)} = 0.01291\ mm^{-1}$. Base heat flow from the reaction at node 1, $R = [K]_{1,*}\{T\} - F_1$: 4.734 W (exact 4.705 W).

Fin parameters: Efficiency of the fin $\eta = Q_{actual} / (h\ P\ L\ (T_b - T_\infty))$ and effectiveness $\varepsilon = Q_{fin}/(h\ A_c(T_b - T_\infty))$. Fins are useful when hP/kA_c is small, i.e. high conductivity and slender.

4.8 More Solved Problems

Problem 5 : strain and stress in a Q4 element

Rectangle 2 × 1 mm : nodes (0,0), (2,0), (2,1), (0,1); plane stress $E = 2\times10^5$, $\nu = 0.3$. Nodes 1, 4 fixed; $u_2 = u_3 = 0.002$ mm, $v_3 = -0.0006$ mm, $v_2 = 0$. At the centre ($\xi = \eta = 0$) :

$[J] = [1\ 0\ ;\ 0\ 0.5]$, $\det J = 0.5$, $J^{-1} = [1\ 0\ ;\ 0\ 2]$. $\partial N_i/\partial\xi = [-1, 1, 1, -1]/4$, $\partial N_i/\partial\eta = [-1, -1, 1, 1]/4 \Rightarrow \partial N_i/\partial x = [-0.25, 0.25, 0.25, -0.25]$, $\partial N_i/\partial y = [-0.5, -0.5, 0.5, 0.5]$.

Strains : $\varepsilon_x = \Sigma\ N_{i,x}u_i = 0.25\times0.002 + 0.25\times0.002 = 0.001000$, $\varepsilon_y = \Sigma\ N_{i,y}v_i = 0.5\times(-0.0006) = -0.000300$, $\gamma_{xy} = \Sigma(N_{i,y}u_i + N_{i,x}v_i) = -0.000150$.

Stress $\{\sigma\} = [D]\{\varepsilon\}$: $\sigma_x = 200.00$, $\sigma_y = 0.00$, $\tau_{xy} = -11.54\ N/mm^2$.

Problem 6 : slab with heat generation

Plate $L = 0.1$ m, $k = 20\ W/m\cdot K$, generation $Q = 2\times10^5\ W/m^3$, both faces held at 100 °C. Two elements, $L_e = 0.05$ m, $A = 1\ m^2$. Conduction $k/L_e = 400\ W/K$.

$$[K]\begin{bmatrix} 400 & -400 & 0 \\ -400 & 800 & -400 \\ 0 & -400 & 400 \end{bmatrix}\begin{Bmatrix} T_1 \\ T_2 \\ T_3 \end{Bmatrix} = [5000, 10000, 5000]^T$$

($F_i = Q\ A\ L_e/2$ per element end). With $T_1 = T_3 = 100$: $800\ T_2 = 10000 + 400(100) + 400(100) \Rightarrow T_2 = 112.5$ °C. Exact maximum $T = T_w + QL^2/(8k) = 100 + 2\times10^5\times0.01/160 = 112.5$ °C (exact at the node).

Problem 7 : same fin with convection at the tip

Add $h\,A_c = 25\times10^{-6}\times78.54 = 1.96\times10^{-3}$ W/K to K_{55} and $hA_cT_\infty = 0.0589$ W to F_5. Solution (°C) :

x (mm)	0	25	50	75	100
T (tip convecting)	120.00	98.97	85.26	77.40	74.56
T (insulated tip)	120.00	99.21	85.76	78.22	75.80

Tip temperature is slightly lower with a convecting tip (extra heat loss), but the difference is small for a slender fin; the exact corrected-length method ($L_c = L + d/4$) predicts the same trend.

Two-dimensional heat conduction (triangle) - outline

Temperature $T = N_1T_1 + N_2T_2 + N_3T_3$ (CST shape functions). Gradient $\{g\} = [\partial T/\partial x,\ \partial T/\partial y]^T = [B]\{T\}$, $B = (1/2A)[b_1\ b_2\ b_3\ ;\ c_1\ c_2\ c_3]$.

$$[K_c] = \int [B]^T\, k\, [B]\, t\, dA = (k\, t/4A)\ (\ \{b\}\{b\}^T + \{c\}\{c\}^T\ )$$

Example : triangle (0,0), (2,0), (0,2), $k = 1$, $t = 1$, $A = 2 \Rightarrow b = [-2,\ 2,\ 0]$, $c = [-2,\ 0,\ 2]$:

$$[K_c] = \begin{bmatrix} 1.00 & -0.50 & -0.50 \\ -0.50 & 0.50 & 0.00 \\ -0.50 & 0.00 & 0.50 \end{bmatrix}$$

Convection on an edge i-j : $[K_h] = (h\ t\ L_{ij}/6)\ [2\ 1\ ;\ 1\ 2]$ and $\{f_h\} = (h\ t\ L_{ij}\ T_\infty/2)\ [1\ 1]^T$. Heat generation : $f = Q\ t\ A/3$ each node. The assembly and BC procedure is identical to the structural case.

Structural ↔ thermal analogy

Structural (bar)	Thermal (1-D conduction)
displacement u	temperature T
force F	heat flow Q
stiffness AE/L	conductance kA/L
spring support	convection hA
body force	heat generation Q

Further remarks on quadrilaterals

- **Reduced integration** (1×1 for Q4) is cheaper but gives zero-energy **hourglass modes**; 2×2 is the standard full integration. Q8 with 2×2 (reduced) is sometimes used to avoid locking.
- A quad can be split into 2 triangles but one gets the CST accuracy; a Q4 represents linear strain variation in part (ε_x varies with y).
- Mesh distortion degrades accuracy; keep interior angles between 45° and 135°, aspect ratio < 5, and avoid concave quadrilaterals (det J changes sign).

Gauss quadrature in 2-D : worked example

Evaluate $I = \int_{-1}^{1}\int_{-1}^{1} (\xi^2\eta^2 + \xi\eta + 1)\, d\xi\, d\eta$. Exact : (2/3)(2/3) + 0 + 4 = 4.4444. 2×2 Gauss (all weights 1, points ±0.57735) :

Point	ξ	η	$f(\xi,\eta)$
1	−0.57735	−0.57735	1.4444
2	0.57735	−0.57735	0.7778
3	0.57735	0.57735	1.4444
4	−0.57735	0.57735	0.7778

Sum = 4.4444 ⇒ I = 4.4444 (exact, since the integrand has degree ≤ 3 in each variable).

Equivalent nodal loads on a Q4 edge

Uniform pressure p on side 2-3 ($\xi = +1$) of length L_{23} = 3 mm, t = 2 mm, normal in +x direction : p = 10 N/mm^2. Along the edge $N_2 = (1-\eta)/2$, $N_3 = (1+\eta)/2$, $ds = (L_{23}/2)d\eta$:

$$f_{2x} = f_{3x} = p\, t\, L_{23}/2 = 10 \times 2 \times 3/2 = 30 \text{ N (y-forces zero)}$$

For a **linearly varying** edge load with $p_2 \to p_3$: $f_2 = tL(2p_2 + p_3)/6$, $f_3 = tL(p_2 + 2p_3)/6$. For the 8-noded element a uniform edge load is shared in the ratio 1 : 4 : 1 (×pLt/6) between corner, mid-side and corner nodes - **not** equally.

Q8 : numerical check of the shape functions

At $(\xi,\eta) = (0.5, 0.5)$: $N_1 = -0.1250$, $N_2 = -0.1875$, $N_3 = 0.0000$, $N_4 = -0.1875$, $N_5 = 0.1875$, $N_6 = 0.5625$, $N_7 = 0.5625$, $N_8 = 0.1875$. Sum = 1.0000 (partition of unity OK). Corner shape functions can be negative inside the element - normal for quadratic elements.

Thermal resistance relations (useful check for FEM results)

Mode	Resistance (per unit area A)	FEM equivalent
Conduction (plane wall)	$R = L/(kA)$	$K = kA/L$
Convection	$R = 1/(hA)$	$K = hA$
Series layers	$R_{tot} = \Sigma R_i$	assembled [K]
Heat flow	$Q = \Delta T/R_{tot}$	$Q = K\,\Delta T$ (series)
Overall U	$1/UA = R_{tot}$	—

Analytical fin results (for comparison with FEM)

Tip condition	Temperature $\theta = T - T_\infty$	Heat loss Q
Insulated tip	$\theta_b \cosh m(L-x)/\cosh mL$	$\sqrt{(hPkA_c)}\,\theta_b \tanh mL$
Convecting tip	$\theta_b [\cosh m(L-x) + (h/mk) \sinh m(L-x)] / [\cosh mL + (h/mk) \sinh mL]$	$\sqrt{(hPkA_c)}\,\theta_b [\sinh mL + (h/mk) \cosh mL]/[\cosh mL + (h/mk) \sinh mL]$
Very long fin	$\theta_b\, e^{-mx}$	$\sqrt{(hPkA_c)}\,\theta_b$

with $m = \sqrt{(hP/kA_c)}$. Efficiency of an insulated-tip fin $\eta = \tanh(mL)/(mL)$.

Example above : $mL = 1.291 \Rightarrow \eta = 0.859/1.291 = 0.665$.

Units: Units : keep one consistent system. With mm, N, W : k in $W/mm \cdot K = (W/m \cdot K)/1000$, h in $W/mm^2K = (W/m^2K)/10^6$. A frequent error is mixing k in $W/m \cdot K$ with lengths in mm.

Short answer questions

Q1 Number of Gauss points for a 4-noded quad ? - $2 \times 2 = 4$. For 8-noded ? - $3 \times 3 = 9$.

Q2 What is the Jacobian ? - The matrix relating derivatives in natural and global coordinates; det J = area scale factor.

Q3 Write N_3 of Q4. - $(1+\xi)(1+\eta)/4$.

Q4 What is the order of exactness of 2-point Gauss rule ? - Polynomials up to degree 3.

Q5. What is heat transfer coefficient h ? - Rate of convective heat transfer per unit area per kelvin, W/m^2K.

Expected questions

- Define isoparametric, sub-parametric, super-parametric elements. Derive shape functions of Q4 / Q8; evaluate Jacobian; derive [k] with Gauss integration.
- Gauss quadrature : points and weights for 2 and 3 points; integrate given function; why 2×2 for Q4 ?
- Derive the 1-D conduction element; solve composite wall; derive element matrices of a fin and find temperature distribution.

UNIT V : Dynamic Analysis

5.1 Torsion of a Uniform Shaft

A circular shaft transmits torque T. Each node has **one dof**, the angle of twist φ (rotation about the shaft axis). A 2-noded element has $\{q\} = [\varphi_1\ \varphi_2]^T$ - the formulation is identical to the 1-D bar element (Unit I) with the analogy :

Axial bar	Torsion of shaft
displacement u	angle of twist φ
axial force F	torque T
E	G (shear modulus)
A	J (polar moment of area)
AE/L	GJ/L

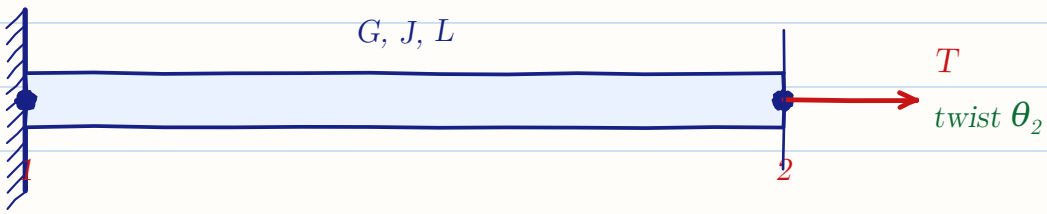


Fig 5.1 Shaft element

Basic relations : $T/J = G\gamma/r = G\, d\varphi/dx \Rightarrow T = G\, J\, d\varphi/dx$. Polar moment : $J = \pi d^4/32$ (solid), $J = \pi(D^4 - d^4)/32$ (hollow). Shear stress $\tau = G\, r\, d\varphi/dx = T\, r/J$; maximum at the outer surface.

Finite element formulation

$$\varphi = N_1\varphi_1 + N_2\varphi_2,\ N_1 = (1-\xi)/2,\ N_2 = (1+\xi)/2\ ;\ d\varphi/dx = (1/L)[-1\ 1]\{\varphi_1\ \varphi_2\}^T$$

Strain energy $U = \tfrac{1}{2} \int G\, J\, (d\varphi/dx)^2\, dx = \tfrac{1}{2}\{q\}^T[k]\{q\}$ and work of the nodal torques $\{T\}$:

$$[k] = GJ/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \varphi_1 \\ \varphi_2 \end{Bmatrix} = [T_1\ T_2]^T$$

Assemble exactly as for bars : node i of an n-element shaft contributes GJ_{i-1}/L_{i-1} + GJ_i/L_i to K_{ii}. Boundary condition : $\varphi = 0$ at a fixed end (elimination/penalty). Torques are applied at nodes; for a distributed torque t per unit length : $\{f\} = tL/2\ [1\ 1]^T$.

Shear stress in an element : $\tau = G\ (d/2)(\varphi_2 - \varphi_1)/L$ (d = diameter of that element).

5.2 Solved Problems on Shafts

Problem 1 : stepped shaft with end torque

Shaft fixed at node 1; element 1 : $d = 40$ mm, $L = 500$ mm; element 2 : $d = 30$ mm, $L = 500$ mm; $G = 8\times10^4$ N/mm^2; torque $T = 10^6$ N · mm at the free end (node 3).

$J_1 = \pi(40)^4/32 = 251327$ mm^4, $J_2 = \pi(30)^4/32 = 79522$ mm^4. $GJ_1/L_1 = 40212386$, $GJ_2/L_2 = 12723450$ N · mm/rad.

$$10^6 \times \begin{bmatrix} 52.936 & -12.723 \\ -12.723 & 12.723 \end{bmatrix} \begin{Bmatrix} \varphi_2 \\ \varphi_3 \end{Bmatrix} = [0,\ 10^6]^T$$

Solution : $\varphi_2 = 0.02487$ rad, $\varphi_3 = 0.10346$ rad (= 5.928 degrees). Shear stresses : $\tau_1 = G(d_1/2)\varphi_2/L_1 = 79.58$ N/mm^2, $\tau_2 = G(d_2/2)(\varphi_3-\varphi_2)/L_2 = 188.63$ N/mm^2.

Check : $\tau = T\,r/J$: element 1 $\rightarrow 10^6\times20/251327 = 79.58$; element 2 $\rightarrow 10^6\times15/79522 = 188.63$ N/mm^2 (same torque flows through both elements).

Problem 2 : shaft fixed at both ends, torque at the middle

Uniform shaft, 2 elements of equal length $L/2$, torque T at node 2, nodes 1 and 3 fixed. Reduced equation : $(2GJ/(L/2)\ldots) = (GJ/L_e)\ [2]\ \varphi_2 = T$ with $L_e = L/2$:

$$\varphi_2 = T\,L_e/(2\,G\,J) = T\,L/(4\,G\,J) \text{ reactions } T_1 = T_3 = T/2$$

Torque in each half = $T/2$ with opposite sense; $\tau_{max} = T\,r/(2J)$.

Problem 3 : three-element shaft with two torques

Uniform shaft $d = 40$ mm, three elements of 300 mm (nodes 1-4), $G = 8\times10^4$ N/mm^2, fixed at node 1; torques 5×10^5 N · mm at node 2 and 3×10^5 N · mm at node 4. $GJ/L = 67020643$ N · mm/rad.

Node	1	2	3	4
φ (rad)	0.00000	0.01194	0.01641	0.02089

Shear stresses : $\tau_1 = 63.66$, $\tau_2 = 23.87$, $\tau_3 = 23.87$ N/mm^2. Reaction torque at node 1 = -800000 N · mm (= $-(5+3)\times10^5$; equilibrium OK).

5.3 Dynamic Analysis : Formulation of the FE Model

For a structure vibrating under time-dependent loads, inertia must be included. Using **Hamilton's principle / Lagrange's equation** with kinetic energy $T = \frac{1}{2} \int \rho \{\dot{u}\}^T \{\dot{u}\} dV$, strain energy $U = \frac{1}{2}\{Q\}^T[K]\{Q\}$, we obtain the equation of motion :

$$[M]\{\ddot{Q}\} + [C]\{\dot{Q}\} + [K]\{Q\} = \{F(t)\}$$

$[M]$ = mass matrix, $[C]$ = damping matrix (Rayleigh : $[C] = a[M] + b[K]$), $[K]$ = stiffness matrix, $\{Q\}$ = nodal displacements. **Free vibration** (no damping, no load) with $Q = \{\varphi\} \sin \omega t$:

$$([K] - \omega^2[M]) \{\varphi\} = \{0\} \Rightarrow \det([K] - \lambda[M]) = 0, \lambda = \omega^2$$

- Eigenvalues $\lambda_i = \omega_i^2$ give natural frequencies ω_i (rad/s), $f_i = \omega_i/2\pi$ (Hz). Number of eigenvalues = number of free dof.
- Eigenvectors $\{\varphi\}_i$ give the **mode shapes**; they are determined up to a scale factor. Normalisation : $\{\varphi\}_i^T[M]\{\varphi\}_i = 1$.
- **Orthogonality** : $\{\varphi\}_i^T[M]\{\varphi\}_j = 0$ and $\{\varphi\}_i^T[K]\{\varphi\}_j = 0$ for $i \neq j$; $\{\varphi\}_i^T[K]\{\varphi\}_i = \omega_i^2$.
- Rigid-body modes of an unrestrained structure have $\omega = 0$ (before applying supports, $[K]$ is singular).

Element mass matrices

From kinetic energy, with displacement $u = [N]\{q\}$:

$$[m] = \int_V \rho\, [N]^T[N]\, dV \text{ (**consistent** mass matrix, same shape functions as stiffness)}$$

$$\text{1-D bar : } [m] = \rho AL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Torsion shaft : } [m] = \rho JL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Derivation (bar) : $[m] = \rho A \int_0^L [N_1\ N_2]^T[N_1\ N_2]\, dx = \rho A\ (L/2) \int_{-1}^{1} [\ N_1^2\ N_1N_2\ ;\ N_1N_2\ N_2^2\ ]\, d\xi$; $\int N_1^2 d\xi \cdot (L/2) = L/3$, $\int N_1N_2\, d\xi \cdot (L/2) = L/6 \Rightarrow \rho AL/6\ [2\ 1\ ;\ 1\ 2]$.

$$\textbf{Lumped} \text{ mass (half the mass at each node) : } [m] = \rho AL/2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Plane truss : } [m] = \rho AL/6 \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$$\text{Beam : } [m] = \rho AL/420 \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$

Remark: *Total mass check : the sum of all entries of the translational part of [m] must equal the element mass ρAL (bar : $(2+1+1+2)\rho AL/6 = \rho AL$). Lumped matrices are diagonal, cheaper, and give a lower-bound (under-estimate) of frequencies, while consistent matrices give an upper bound.*

(lumped model of a bar / shaft)

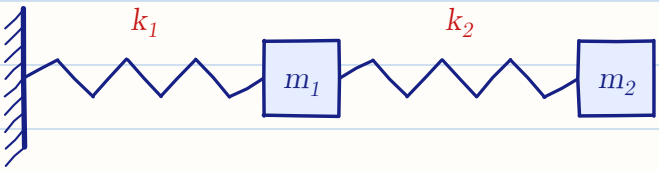


Fig 5.2 Lumped spring-mass equivalent of a bar

5.4 Eigenvalues and Eigenvectors : Bar and Shaft

Procedure

1. *Find [k] and [m] for each element; assemble [K], [M].*
2. *Apply boundary conditions (delete rows/columns of restrained dof) - **both** [K] and [M].*
3. *Solve the characteristic equation $\det([K] - \lambda[M]) = 0$ for $\lambda_i = \omega_i^2$.*
4. *Substitute each λ_i in $([K] - \lambda_i[M])\{\varphi\} = 0$, fix one component = 1 and find the others (mode shape).*
5. *Normalise the modes if required and check orthogonality.*

Problem 4 : fixed-free bar, one element

Uniform bar fixed at node 1, free at node 2 (L, A, E, ρ). Node 2 only : $K = AE/L$, $M = \rho AL/3$ (consistent) or $\rho AL/2$ (lumped).

Consistent : $\omega^2 = 3E/(\rho L^2) \Rightarrow \omega = 1.732\ \sqrt{(E/\rho)}/L$ Lumped : $\omega = 1.414\ \sqrt{(E/\rho)}/L$ Exact : $1.571\ \sqrt{(E/\rho)}/L$

For steel ($E = 2\times10^5$ N/mm^2, $\rho = 7.85\times10^{-9}$ N·s^2/mm^4), $L = 1000$ mm : $\sqrt{(E/\rho)}/L = 5047.5$ rad/s $\Rightarrow$ exact $\omega_1 = 1.5708\times5047.5 = 7929$ rad/s (1262 Hz); consistent 1-element = 8742 rad/s; lumped = 7137 rad/s.

Problem 5 : fixed-free bar, two elements (consistent mass)

Let $L_e = L/2$. Reduced matrices (nodes 2, 3) : $[K] = (AE/L_e)\ [2\ -1\ ;\ -1\ 1]$, $[M] = (\rho AL_e/6)\ [4\ 1\ ;\ 1\ 2]$.

$\det([K] - \lambda[M]) = 0$ with $\Lambda = \lambda\ \rho\ L_e^2/(6E)$: $(2 - 4\Lambda)(1 - 2\Lambda) - (-1 - \Lambda)^2 = 0$ $\Rightarrow 7\Lambda^2 - 10\Lambda + 1 = 0 \Rightarrow \Lambda = 0.1082,\ 1.3203.$

$\omega_1 = \sqrt{(6\times0.1082)}\ \sqrt{(E/\rho)}/L_e = 0.806\ \sqrt{(E/\rho)}/L_e = 1.611\ \sqrt{(E/\rho)}/L$ $\omega_2 = 2.815\ \sqrt{(E/\rho)}/L_e = 5.629\ \sqrt{(E/\rho)}/L$

Mode	Exact ($\pi(2n-1)/2$)	1 elem (cons.)	2 elem (cons.)	Error (2 elem)
1	1.5708	1.7321	1.6113	2.6 %
2	4.7124	–	5.6293	19.5 %

Mode shapes (normalised to the tip = 1) : substituting Λ_1 in $(2-4\Lambda)\varphi_2 - (1+\Lambda)\varphi_3 = 0$: $\{\varphi\}_1 = [0.7071\ ,\ 1]^T$ (node 2, node 3), $\{\varphi\}_2 = [-0.7071\ ,\ 1]^T$. The first mode has no sign change; the second has one node (sign change) - matches $\sin(\pi x/2L)$ and $\sin(3\pi x/2L)$.

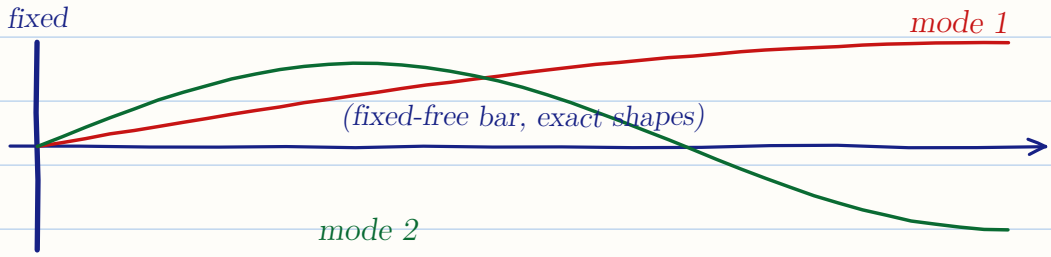


Fig 5.3 First two modes of the fixed-free bar

Orthogonality check : $\{\varphi\}_1^T[M]\{\varphi\}_2 = 0.00000 \approx 0$ OK.

5.5 Solved Problems : Shaft Vibration

Problem 6 : torsional vibration of a fixed-free shaft

Shaft $L = 1000$ mm, $d = 50$ mm, $G = 8\times10^4$ N/mm^2, $\rho = 7.85\times10^{-9}$ N $\cdot$ s^2/mm^4.

Two elements. The matrices are the bar matrices with ($AE \to GJ$, $\rho A \to \rho J$), so :

$$[K] = GJ/L_e \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$[M] = \rho J L_e/6 \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$$

$\omega_1 = 1.611\ \sqrt{(G/\rho)}/L = 1.611 \times 3192 = 5144$ rad/s (818.7 Hz); $\omega_2 = 5.629 \times 3192 = 17971$ rad/s. Exact : $\omega_1 = (\pi/2)\sqrt{(G/\rho)}/L = 5015$ rad/s.

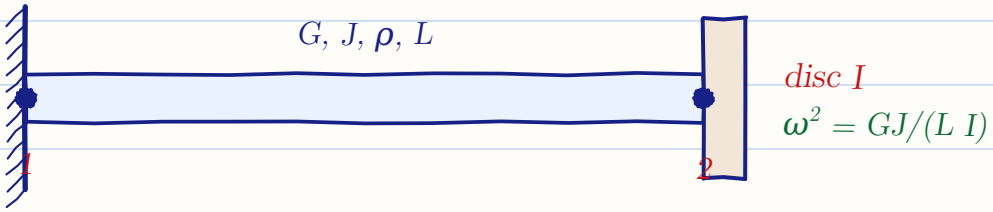


Fig 5.4 Shaft with a disc at the free end

Problem 7 : shaft carrying a disc (massless shaft, one element)

Shaft $d = 30$ mm, $L = 500$ mm, $G = 8\times10^4$ N/mm^2; disc of polar mass moment of inertia $I = 0.05$ kg $\cdot$ m^2 $= 50$ N $\cdot$ mm $\cdot$ s^2. With one dof φ_2 : $K = GJ/L = 12723450$ N $\cdot$ mm/rad, $M = I = 50$:

$$\omega^2 = GJ/(L\ I) = 254469 \Rightarrow \omega = 504.4 \text{ rad/s } (f = 80.3 \text{ Hz})$$

If shaft inertia is to be included, add $\rho JL/3$ (consistent) to the disc inertia at the free node : $M = I + \rho JL/3$ (small here : $\rho JL/3 = 0.10404$ N $\cdot$ mm $\cdot$ s^2), so ω decreases very slightly.

Problem 8 : free-free bar (unrestrained) - rigid body mode

One element with both nodes free : $\det([K] - \lambda[M]) = 0$ with $[K] = (AE/L)[1\ -1\ ;\ -1\ 1]$, $[M] = (\rho AL/6)[2\ 1\ ;\ 1\ 2]$:

$$\lambda_1 = 0 \text{ (rigid-body translation) } \lambda_2 = 12E/(\rho L^2) \Rightarrow \omega_2 = 3.464\ \sqrt{(E/\rho)}/L \text{ (exact } \pi = 3.142)$$

Eigenvectors : $\{\varphi\}_1 = [1,\ 1]^T$ (both nodes move together), $\{\varphi\}_2 = [1,\ -1]^T$ (nodes move oppositely). λ_2 computed numerically $= 12.0000 \times E/(\rho L^2)$ OK.

Problem 9 : cantilever beam frequency (1 element)

With the beam matrices (Unit II and above), deleting v_1, θ_1 : $(EI/L^3)[12\ -6L\ ;\ -6L\ 4L^2]$, $(\rho AL/420)[156\ -22L\ ;\ -22L\ 4L^2]$. Solving gives

$\omega_1 = 3.533\ \sqrt{(EI/\rho AL^4)} = 3.533\ \sqrt{(EI/\rho A)}/L^2$ (exact 3.516) $\omega_2 = 34.81\ \sqrt{(EI/\rho AL^4)}$ (exact 22.03)

Observation: A single cubic Hermite element gives the first frequency within 0.5 % but the second mode is poor (about 58 % high). More elements are required for higher modes: rule of thumb - use about 2 elements per half-wave of the highest mode of interest.

5.6 Derivation of the Dynamic Equations

Lagrange's equation for a conservative system with generalised coordinates Q_i :

$d/dt\ (\partial T/\partial \dot{Q}_i) - \partial T/\partial Q_i + \partial U/\partial Q_i = F_i(t)$.

1. Kinetic energy of an element : $T_e = \tfrac{1}{2} \int \rho\ \{\dot{u}\}^T\{\dot{u}\}\ dV$, with $\{\dot{u}\} = [N]\{\dot{q}\} \Rightarrow$ $T_e = \tfrac{1}{2}\ \{\dot{q}\}^T\ (\int \rho\ [N]^T[N]\ dV\ )\ \{\dot{q}\} = \tfrac{1}{2}\ \{\dot{q}\}^T[m]\{\dot{q}\}$.
2. Strain energy $U_e = \tfrac{1}{2}\ \{q\}^T[k]\{q\}$. Sum over elements : $T = \tfrac{1}{2}\ \{\dot{Q}\}^T[M]\{\dot{Q}\}$, $U = \tfrac{1}{2}\ \{Q\}^T[K]\{Q\}$.
3. Substitute in Lagrange's equation : $\partial T/\partial \dot{Q} = [M]\{\dot{Q}\}$, d/dt gives $[M]\{\ddot{Q}\}$; $\partial U/\partial Q = [K]\{Q\}$. Hence $[M]\{\ddot{Q}\} + [K]\{Q\} = \{F(t)\}$.
4. Damping adds $[C]\{\dot{Q}\}$ (dissipation function $\tfrac{1}{2}\{\dot{Q}\}^T[C]\{\dot{Q}\}$).

Consistent mass matrix of the beam element (outline)

$[m] = \rho A \int_0^L [H]^T[H]\ dx = \rho A\ (L/2) \int_{-1}^{1} [H]^T[H]\ d\xi$, with the Hermite cubics of Unit II. Example : $m_{11} = \rho A\ (L/2)(1/16) \int\ (1-\xi)^4(2+\xi)^2\ d\xi = \rho A\ L\ (156/420)$; $m_{22} = \rho A\ (L/2)(L^2/64) \int\ (1-\xi)^4(1+\xi)^2\ d\xi = \rho A\ L^3(4/420)$.

Forced vibration (harmonic) in brief

For $F(t) = F_0 \sin \Omega t$ the steady response is $\{Q\} = \{Q_0\} \sin \Omega t$ with $(\ [K] - \Omega^2[M]\ )\{Q_0\} = \{F_0\}$. When Ω approaches a natural frequency ω_i the matrix becomes singular - **resonance**. With damping the amplitude of mode i is limited by $1/(2\zeta_i)$.

5.7 More Solved Problems

Problem 10 : convergence of bar frequencies with mesh refinement

Fixed-free bar. Frequencies are listed as the coefficient C in $\omega = C\sqrt{(E/\rho)}/L$. Consistent mass gives upper bounds, lumped mass gives lower bounds, and both converge to the exact values :

Elements	Consistent ω_1	Consistent ω_2	Consistent ω_3	Lumped ω_1	Lumped ω_2
1	1.7321	–	–	1.4142	–
2	1.6114	5.6293	–	1.5307	3.6955
3	1.5888	5.1962	9.4266	1.5529	4.2426
4	1.5809	4.9872	9.0594	1.5607	4.4446
6	1.5753	4.8342	8.4188	1.5663	4.5922
Exact	1.5708	4.7124	7.8540	1.5708	4.7124

Observation : the lower modes converge fast; the 3rd mode needs at least 4-6 elements. The **average** of consistent and lumped results is often very close to the exact value.

Problem 11 : fixed-fixed bar, three elements

Interior nodes 2, 3 free (4 nodes, ends fixed); $[K] = (AE/L_e)[2\ -1\ ;\ -1\ 2]$, $[M] = (\rho AL_e/6)[4\ 1\ ;\ 1\ 4] \Rightarrow \det = 0$ gives $(2-4\Lambda)^2 - (1+\Lambda)^2 = 0 \Rightarrow \Lambda = 1/5,\ 1 \Rightarrow \omega = \sqrt{(6\Lambda)}/L_e$. With $L_e = L/3$:

Mode	FEM (3 elem, consistent)	Exact nπ	Error %
1	3.2863	3.1416	4.61
2	7.3485	6.2832	16.95

Mode shapes : mode 1 $\{\varphi\} = [1,\ 1]^T$ (symmetric, both interior nodes move the same way); mode 2 $\{\varphi\} = [1,\ -1]^T$ (antisymmetric, a node at the centre).

Problem 12 : two-disc shaft (torsional vibration)

The stepped shaft of Problem 1 ($k_1 = 40212386$, $k_2 = 12723450$ N · mm/rad) is fixed at node 1 and carries discs $I_2 = 100$ at node 2 and $I_3 = 50$ N · mm · s^2 at node 3 (shaft mass neglected, so $[M]$ is diagonal).

$$[K]\ (\times 10^6) = \begin{bmatrix} 52.936 & -12.723 \\ -12.723 & 12.723 \end{bmatrix}$$

Frequency equation : $\det([K] - \omega^2[M]) = 0 \Rightarrow \omega_1 = 406.8$ rad/s, $\omega_2 = 786.3$ rad/s

(f = 64.7, 125.2 Hz).

Mode shapes (disc 2 = 1) : $\{\varphi\}_1 = [1,\ 2.860]^T$ (both discs rotate in the same sense), $\{\varphi\}_2 = [1,\ -1.414]^T$ (opposite senses - one node on the shaft between the discs).

Problem 13 : two-dof spring-mass (lumped bar) - hand solution

Fixed-free bar of two equal elements with lumped masses m at node 2 and m at node 3 (end mass taken = m for simplicity); $k = AE/L_e$. $[K] = k[2\ -1\ ;\ -1\ 1]$, $[M] = m\ [1\ 0\ ;\ 0\ 1]$.

$(2k - \lambda m)(k - \lambda m) - k^2 = 0 \Rightarrow \lambda^2 m^2 - 3km\lambda + k^2 = 0 \Rightarrow \lambda = (k/m)(3 \mp \sqrt{5})/2 = 0.382\ k/m\ ,\ 2.618\ k/m$

$\omega_1 = 0.618\ \sqrt{(k/m)}$, $\omega_2 = 1.618\ \sqrt{(k/m)}$. Mode shapes : from $(2k - \lambda m)\varphi_2 - k\varphi_3 = 0$: $\lambda_1 \Rightarrow \varphi_3/\varphi_2 = 2 - 0.382 = 1.618 \Rightarrow \{\varphi\}_1 = [1,\ 1.618]^T$; $\lambda_2 \Rightarrow \varphi_3/\varphi_2 = 2 - 2.618 = -0.618 \Rightarrow \{\varphi\}_2 = [1,\ -0.618]^T$.

Orthogonality with $[M] = mI$: $1\times1 + 1.618\times(-0.618) = 1 - 1.000 = 0$ OK.

Natural frequency checklist

- Convert units first : $\rho = 7850\ kg/m^3 = 7.85\times10^{-9}\ N\cdot s^2/mm^4$ (in N-mm-s system) ; $E = 200\ GPa = 2\times10^5\ N/mm^2$; $G = 80\ GPa = 8\times10^4\ N/mm^2$.
- Use the **same** BC for $[K]$ and $[M]$. ω in rad/s ; $f = \omega/2\pi$; $T = 1/f$.
- Number of modes = number of free dof; FE frequencies from a consistent mass are upper bounds of the exact ones.

Time response and damping (brief)

- Modal superposition : $\{Q\} = \Sigma\ \{\varphi\}_i\ \eta_i(t)$; the equations decouple to $\ddot{\eta}_i + 2\zeta_i\omega_i\ \dot{\eta}_i + \omega_i^2\eta_i = \{\varphi\}_i^T\{F(t)\}$. Direct methods : central difference, Newmark-β.
- Numerical eigen-solvers : characteristic polynomial (small systems), inverse (power) iteration, Jacobi, subspace iteration, Lanczos (large systems).
- Rayleigh quotient $\omega^2 \approx \{\varphi\}^T[K]\{\varphi\}/\{\varphi\}^T[M]\{\varphi\}$ gives an upper bound to ω_1^2 for any trial vector.

Summary of mass and stiffness matrices

Element	[k]	[m] consistent
Bar	AE/L [1 −1 ; −1 1]	$\rho AL/6$ [2 1 ; 1 2]
Torsion	GJ/L [1 −1 ; −1 1]	$\rho JL/6$ [2 1 ; 1 2]
Beam	EI/L^3[12 6L ...]	$\rho AL/420$ [156 22L ...]
CST (plane)	$tA\ B^TDB$	$\rho tA/12$ [2 0 1 0 1 0 ; ...]

5.8 Further Examples and Theory

Problem 14 : cantilever beam, convergence with elements

Cantilever beam of uniform section; the results are the coefficient C in $\omega = C\sqrt{(EI/\rho A)/L^2}$ (consistent mass, Hermite elements) :

Elements	ω_1	ω_2	ω_3
1	3.533	34.81	–
2	3.518	22.22	75.16
4	3.516	22.06	62.17
Exact	3.516	22.03	61.70

Two elements already give ω_1 to 4 digits; higher modes converge more slowly - a typical property of FE eigenvalue solutions.

Problem 15 : Rayleigh quotient estimate

For the fixed-free bar take the trial shape $u = 2x/L - x^2/L^2$ (satisfies $u(0) = 0$, and $u'(L) = 0$).

$$\omega^2 \approx \int AE\,(u')^2 dx \,/\, \int \rho A\, u^2\, dx = [4AE/(3L)] \,/\, [\rho AL\ (8/15)] = 2.5\ E/(\rho L^2) \Rightarrow \omega = 1.581\sqrt{(E/\rho)}/L$$

Exact = 1.5708; the estimate is an **upper bound** (error 0.7 %). A trial $u = x/L$ gives $\omega = 1.732$, i.e. the one-element FE value. Rayleigh quotient is stationary, so a first-order error in the shape produces only a second-order error in ω^2.

Problem 16 : bar with an end mass

Steel bar ($A = 100\ mm^2$, $L = 500\ mm$, $E = 2\times10^5$) fixed at one end, carrying a lumped mass $m = 2\ kg = 2\times10^{-3}\ N\cdot s^2/mm$ at the free end; bar mass neglected, one element : $K = AE/L = 4\times10^4\ N/mm$.

$$\omega = \sqrt{(K/m)} = \sqrt{(4\times10^4/2\times10^{-3})} = 4472.1\ rad/s\ (f = 711.8\ Hz)$$

Adding the bar mass (consistent) : $M = m + \rho AL/3 = 2\times10^{-3} + 0.000131 = 0.002131 \Rightarrow \omega = 4332.7$ rad/s (slightly lower - the bar mass is only 19.63 % of the end mass).

Mass matrices of 2-D elements (for reference)

$$CST:\ [m] = \rho tA/12 \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 \end{bmatrix}$$

- Lumped CST mass : $\rho tA/3$ at each translational dof. Q4 : $[m] = \rho\ t \int\int [N]^T[N]\ \det J\ d\xi\ d\eta$ (2×2 Gauss). Axisymmetric ring : replace tA by $2\pi\ \bar{r}\ A$.
- Row-sum lumping : replace each row of the consistent matrix by its sum on the diagonal. For the bar : $\rho AL/6(2+1) = \rho AL/2$ OK.
- HRZ lumping is used for higher-order elements to avoid negative masses (Q8 corner nodes).

Damping

Rayleigh damping $[C] = \alpha[M] + \beta[K]$. Modal damping ratio $\zeta_i = \alpha/(2\omega_i) + \beta\omega_i/2$. Given ζ at two frequencies ω_1, ω_2 : $\alpha = 2\omega_1\omega_2(\zeta_1\omega_2 - \zeta_2\omega_1)/({\omega_2}^2 - {\omega_1}^2)$, $\beta = 2(\zeta_2\omega_2 - \zeta_1\omega_1)/({\omega_2}^2 - {\omega_1}^2)$.

Direct time integration (reference)

Method	Formula	Nature
Central difference	$Q_{n+1} = 2Q_n - Q_{n-1} + \Delta t^2\ M^{-1}(F - KQ)_n$	explicit, conditionally stable ($\Delta t < 2/\omega_{max}$)
Newmark-β (β=1/4, γ=1/2)	average acceleration	implicit, unconditionally stable

Short answer questions

Q1. What is a consistent mass matrix ? - One derived with the same shape functions used for stiffness : $[m] = \int \rho [N]^T [N] dV$.

Q2. Write the equation of free vibration. - $([K] - \omega^2 [M])\{\varphi\} = 0$.

Q3. What is the analogy between bar and shaft ? - $u \rightarrow \varphi$, $AE \rightarrow GJ$, $\rho A \rightarrow \rho J$.

Q4. Why is [M] of a bar not diagonal in the consistent form ? - Because $N_1 N_2 \neq 0$ couples the two nodes.

Q5. Units of ρ in the N-mm system ? - $N \cdot s^2/mm^4$ (steel 7.85×10^{-9}).

Q6. What are the physical meanings of eigenvalue and eigenvector ? - ω^2 and mode shape (relative amplitudes of dof in free vibration).

Expected questions

- Derive the stiffness matrix for a shaft element; solve stepped/uniform shaft for twist and shear stress.
- Derive consistent and lumped mass matrices of a bar; write the eigenvalue problem.
- Find natural frequencies and mode shapes of a fixed-free bar / shaft with 1 or 2 elements; check orthogonality.
- Write the equation of motion for the dynamic analysis; what are modal analysis and damping ?